

Policy-Relevant Block Abstraction for Sequential Farmland Consolidation Scenario Screening

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Abstract

Farmland consolidation policy increasingly requires planners to compare candidate project areas before field investigation, ownership negotiation, and statutory approval. Such screening is difficult because irregular cadastral parcels, terrain, physical barriers, and threshold-based field-block targets interact. This paper evaluates a policy-relevant block abstraction for sequential farmland-consolidation scenario screening. We aggregate swappable farmland and forest parcels into barrier-aware consolidation blocks, then combine a learned block scheduler with a deterministic connectivity-aware execution engine that performs parcel-level farmland–forest exchanges inside selected blocks. The reward and evaluation metrics combine slope reduction, farmland contiguity, and *baimu fang* formation (100-mu contiguous operating fields), treating *baimu fang* as a local policy instance of minimum contiguous management-unit provision. Experiments use three townships in Bishan District, Chongqing, derived from 3,607–12,950 Third National Land Survey parcels and 78–338 constructed blocks. The block-level formulation trains stably across all 15 seed–township combinations. The clearest scheduler-specific result occurs on the medium township, where the learned scheduler is the only tested block-selection rule, including reward-aware one-step and finite-depth lookahead baselines, to produce positive *baimu fang* area growth. On the small and large townships, reward-aware greedy planning remains competitive or stronger on several metrics. The contribution is therefore not a general claim that reinforcement learning dominates heuristics, but a planning-scale abstraction that makes consolidation trade-offs comparable and shows where learned scheduling can add value under multi-objective ten-

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sion. Area-balance and compactness audits further show that the outputs are opportunity signals, not implementation-ready land-adjustment plans.

Keywords: land use policy, farmland consolidation, spatial abstraction, cadastral parcels, deep reinforcement learning, geospatial decision support, baimu fang

1. Introduction

Farmland consolidation—the spatial reorganization of fragmented agricultural parcels into contiguous, productive blocks—is a recurring land-use policy instrument for improving agricultural production conditions, coordinating ecological restoration, and implementing rural territorial planning. In China, it is central to Comprehensive Land Consolidation policy (*quanyu tudi zonghe zhengzhi*) and high-standard farmland construction. The national standard GB/T 30600-2022 “General Rules for High-Standard Farmland Construction” [8] defines minimum field-block size, slope, contiguity, and access requirements for recognized consolidated fields, while the National High-Standard Farmland Construction Plan (2021–2030) [14] sets quantitative provincial targets. In project practice, the *baimu fang* (100-mu consolidated field, ≥ 6.67 ha of contiguous farmland) and *qianmu fang* (1,000-mu field, ≥ 66.7 ha) serve as operational planning units, reflecting the policy demand for large, connected arable blocks rather than dispersed smallholdings [13, 12]. For readers outside China, *baimu fang* can be read as one local instance of a broader planning problem: providing minimum contiguous management units large enough to support mechanized operation, infrastructure investment, or ecological restoration.

The immediate planning challenge is not to automate final parcel exchanges. Consolidation authorities first need to screen candidate villages or project areas, compare objective trade-offs under fixed budgets, and decide which locations merit field investigation, ownership verification, household consultation, and engineering design. In mountainous southwest China, this screening stage is difficult because steep slopes, fragmented ownership, interspersed forest, roads, waterways, and policy thresholds interact at cadastral scale. A physically attractive parcel exchange may fail later because it disrupts a qualifying field block, violates farmland-area balance, creates an elongated field, or crosses ownership and village-group boundaries that raise negotiation costs.

This paper develops a decision-support method for that early screening problem. Deep reinforcement learning (DRL) has become a useful paradigm for sequential spatial decision problems, including urban configuration and ecological connectivity planning, but its value in cadastral farmland consolidation remains uncertain. Real cadastral parcels differ from regular grid cells in area, shape, neighborhood density, and barrier structure. These properties change the optimization problem rather than merely increasing its size. On irregular cadastral parcels, individual farmland–forest exchanges can approach *swap near-independence*: parcel areas are highly heterogeneous, Queen-contiguity graphs are sparse, and fixed barriers split the swappable graph into small components. We treat this as a design diagnosis, not as a definitive negative benchmark for all parcel-level DRL formulations.

The diagnosis motivates a coarser decision scale that is closer to policy practice. Instead of selecting individual parcels, a screening tool can allocate a fixed consolidation budget across spatial units that compete for resources. Selecting one unit consumes budget that cannot be invested elsewhere. This resource-competition structure creates sequential dependencies while matching the way land consolidation projects are usually organized around project areas, villages, or field blocks. The resulting question is policy-facing: can a block-level abstraction make alternative consolidation scenarios tractable, comparable, and interpretable before the statutory planning process begins?

We propose a **policy-relevant block abstraction** for sequential farmland-consolidation scenario screening. The framework consists of three stages:

1. **Spatial restructuring:** Thousands of irregular parcels are aggregated into tens to hundreds of spatially coherent *consolidation blocks* using a hybrid method combining natural barrier segmentation (roads, waterways, construction land) with constrained agglomerative clustering.
2. **Block-level scheduling:** The optimization is formulated as a Markov Decision Process where the agent selects which block to invest consolidation resources in at each step, observing per-block characteristics and global state metrics.
3. **Connectivity-aware execution:** Within each selected block, a deterministic greedy engine executes farmland–forest swaps using a novel connectivity-aware scoring function that preferentially removes steep, isolated farmland and converts gentle, farmland-adjacent forest—creating spatial spillover effects across block boundaries.

This “*macroscopic planning, microscopic execution*” design draws an analogy to real-world consolidation governance: planning agencies allocate annual consolidation quotas across townships and villages, while local implementers evaluate specific parcel-level changes within assigned areas. Our framework encodes this two-level decision structure, making the block sequence interpretable as a candidate-area prioritization rather than as a binding parcel-exchange prescription. Fig. 1 illustrates the overall architecture.

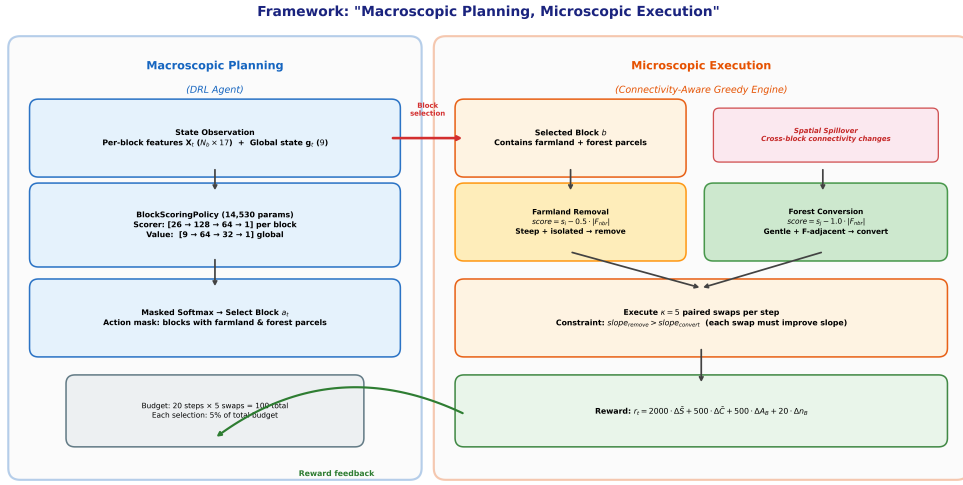


Figure 1: “Macroscopic planning, microscopic execution” framework architecture. The DRL agent allocates a consolidation budget across competing blocks (left), while a deterministic greedy engine executes optimized parcel swaps within each selected block (right). Spatial spillover across block boundaries creates the sequential dependencies that DRL exploits.

A key policy link is the introduction of *baimu fang* formation as both a reward signal and an evaluation metric. In China, *baimu fang* count and area are actionable planning targets: consolidation projects are evaluated and funded partly through the number and total area of newly formed consolidated fields. For readers outside this policy setting, the same construct can be replaced by locally meaningful thresholds, such as minimum mechanized-field size, contiguous management-unit area, or minimum restoration-patch area. By incorporating this metric, the framework translates a planning threshold into a computable scenario indicator.

The intended decision role is deliberately narrow. Table 1 positions the

model in a land-consolidation workflow: it supports pre-project screening, candidate-block ranking, and scenario comparison before field investigation and village negotiation. It does not replace statutory farmland-area balance checks, household consent, compensation design, engineering acceptance, or final administrative approval.

Table 1: Policy workflow and use boundary of the proposed framework. The model produces scenario indicators for planning discussion, not binding parcel-exchange decisions.

Planning stage	Inputs	Model output	Human decision not automated
Project-bank screening	Restricted parcel map, slope, land-use class, fixed barriers, policy thresholds	Ranked candidate blocks and township-level trade-off indicators	Whether a township or village should enter the formal project bank
Scenario comparison	Budget, reward weights, block sequence, baseline alternatives	Slope, contiguity, <i>baimu fang</i> count/area, area-drift diagnostics	Which policy objective should dominate under local planning priorities
Pre-negotiation review	Candidate block sequence and parcel-level exchange trace	Locations where physical gains appear feasible	Ownership verification, household consent, compensation, and village-group coordination
Final check	Area balance, engineering standards, field roads, irrigation, compactness, ecological restrictions	Diagnostic flags and stress-test results	Final land-adjustment plan, acceptance, and statutory approval

We evaluate the framework on three townships spanning a $3.6\times$ range in parcel count (3,607 to 12,950), comparing against five block-selection baselines that share the same within-block engine (Greedy-Global, Greedy-

Sequential, Random-Block, Round-Robin, and a one-step Reward-Greedy lookahead that has access to the same multi-objective reward DRL is trained on; Section 4.7) across 5 random seeds per township. The principal findings are:

1. The block-level MDP trains stably in all 15 seed–township combinations under the operational convergence definition used here, with no instances of degenerate or bimodal seed behavior under the tested settings.
2. Under identical hyperparameters, the learned scheduler produces township-specific block sequences: a moderate, balanced profile on the small township (Township A); a contiguity-and-*baimu fang*-area-retaining profile on the medium township (Township B); and a *baimu fang* count-focused profile on the large township (Township C). These profiles are induced by local spatial structure rather than tuned per township.
3. On Township B, the learned scheduler is the only tested block-selection rule—including reward-aware one-step, depth-2, and depth-3 lookahead baselines that share its reward function—to achieve a positive *baimu fang* area outcome (+12.9 ha vs. Reward-Greedy’s −101.9 ha, depth-2 lookahead’s −108.8 ha, and depth-3 beam lookahead’s −103.8 ha). This is the strongest evidence in the present experiments that learned scheduling contributes a multi-step planning increment on top of access to the multi-objective signal.
4. On Township A and C, Reward-Greedy is competitive with or slightly exceeds DRL on three of four metrics, indicating that block-level abstraction itself, rather than RL-induced multi-step planning, drives much of the multi-objective gain we report. We therefore frame the contribution as “policy-relevant block abstraction with a learnable scheduler,” not as a blanket claim that reinforcement learning beats heuristics.
5. Policy-risk checks clarify the implementation boundary. The learned histories have small farmland-area drift in the original engine (Township B: $+0.28 \pm 0.12\%$), but replay under explicit area-tolerance and exact area-weighted slope checks no longer preserves the positive *baimu fang* area outcome. Compactness auditing further shows that topological *baimu fang* counts are upper-bound estimates of engineering-qualifying fields. The current output is therefore a scenario-screening

indicator, not an area-balanced implementation plan.

These results support spatial abstraction as a practical design choice for policy-facing scenario exploration in farmland consolidation, while leaving direct parcel-level DRL benchmarking, fully retrained area-constrained policies, and cross-district validation as important robustness checks. The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the study area and data; Section 4 presents the methodology in detail; Section 5 reports experimental results; Section 6 discusses implications and limitations; and Section 7 concludes.

2. Related Work

2.1. Farmland consolidation optimization

Farmland consolidation has been studied extensively using operations research and spatial optimization methods. Multi-objective evolutionary algorithms, particularly NSGA-II [6] with spatial extensions [3] and boundary-based genetic algorithms [4], have been applied to land-use allocation at village and county scales. Hybrid genetic-game-theoretic approaches have been used for multi-objective land-use spatial optimization [11], and ant-colony / cellular-automata couplings have been explored for land-use allocation [10]. Integer programming approaches can guarantee optimality for linearizable objectives but struggle with neighborhood-dependent spatial metrics [1]. However, these methods share a common limitation: they treat optimization as a static assignment problem, solving for a complete configuration without exploiting the sequential structure of consolidation decisions.

Recent work in China has emphasized the formation of high-standard farmland blocks (*gaobiaozhun nongtian*) as the operational unit for consolidation planning. The national standard GB/T 30600-2022 [8] specifies minimum slope, field size, contiguity, road access, and irrigation requirements, while the National High-Standard Farmland Construction Plan (2021–2030) [14] sets provincial quantitative targets. The *baimu fang* (100-mu field, ≥ 6.67 ha) serves as the fundamental planning unit in many regional projects, with explicit numeric targets specified in consolidation project approval documents. Earlier empirical work in agricultural economics has documented productivity gains from land consolidation at the household level [17]. Despite this policy emphasis, existing optimization methods operate at the parcel level without explicit consideration of consolidated block formation.

2.2. DRL for spatial optimization

Deep reinforcement learning has been applied to several spatial planning tasks. Zheng et al. [18] applied DRL to spatial planning of urban communities, demonstrating that policy-gradient methods can learn effective spatial configurations. Wang et al. [16] applied adversarial learning to urban generation. In ecological planning, Equihua et al. [7] applied DRL to connectivity conservation planning. These studies show that learned policies can support spatial configuration problems, but they do not address rural cadastral consolidation where thousands of irregular parcels, hard land-type constraints, and policy-defined field-block targets interact.

The gap addressed here is therefore narrower than generic spatial DRL. We ask whether a block-level allocation problem, coupled with deterministic within-block execution, can recover useful sequential structure for farmland consolidation on real cadastral data. The claim is made and tested entirely within this manuscript: the motivation, method, baselines, ablations, and limitations are all reported here, so the contribution can be evaluated without consulting any separate manuscript.

Within the published literature reviewed for this study, the unresolved gap is the combination of block-level sequential allocation, deterministic parcel-level execution, and policy-relevant *baimu fang* metrics for cadastral farmland consolidation.

2.3. Hierarchical and multi-scale optimization

The idea of decomposing spatial optimization across scales has precedent in operations research. Hierarchical reinforcement learning (HRL) decomposes complex tasks into sub-goals at different levels of abstraction [2]. In spatial planning, Li et al. [10] proposed coupling different optimization scales through cellular automata. However, formal HRL methods (e.g., options framework, feudal networks) introduce substantial algorithmic complexity and training instability. Our approach achieves a similar scale decomposition through a simpler mechanism: the block-level agent selects *where* to invest, while a deterministic engine handles *how* to execute—avoiding the need to learn policies at both levels simultaneously.

2.4. Multi-objective and constrained reinforcement learning

The reward signal in this paper combines four channels (slope, contiguity, *baimu fang* area, *baimu fang* count) and is treated as a weighted scalar. This is the simplest of the multi-objective RL approaches and corresponds to a

single point on the Pareto front determined by the chosen weights. More sophisticated treatments include vector-valued value functions and Pareto-conditioned policies in multi-objective RL, and Lagrangian dual methods and constrained policy optimization in constrained MDPs. We adopt the scalar-weighted formulation here because the weights have a direct policy interpretation (slope reduction is the primary objective, *baimu fang* formation is the policy-relevant secondary objective) and because the Pareto-front exploration we report (Section 5.6) is sufficient to support the qualitative claim that block-level DRL discovers trade-offs not reachable by single-objective baselines. A constrained-MDP variant in which farmland area conservation and *baimu fang* qualification thresholds are formulated as hard constraints, with slope reduction as the unconstrained maximand, is a natural extension that we discuss in the limitations section.

2.5. Spatial aggregation and constrained regionalization

The block construction step (Section 4.2) is closely related to a body of GIScience methods for partitioning a set of spatial units into a smaller number of contiguous regions subject to constraints—commonly referred to as constrained regionalization, zoning, or clustering with contiguity constraints. Algorithms such as SKATER, REDCAP, and constrained agglomerative clustering with a spatial connectivity matrix (the latter is what we use, via the `libpysal` / `scikit-learn` pipeline) produce contiguous regions that respect a user-supplied adjacency graph. We use the natural-barrier graph—roads, waterways, construction land excluded—as that adjacency, which embeds physical landscape barriers in the regionalization step itself. This makes our blocks fully consistent with the way Chinese consolidation projects partition the rural landscape (i.e., barriers segment the project area before clustering takes over within each barrier-delimited piece), and it gives the resulting decision units a direct physical interpretation rather than a purely statistical one.

3. Study Area and Data

3.1. Study area

The study area is Bishan District (29.4°–29.7°N, 106.0°–106.3°E), Chongqing Municipality, southwest China. Bishan lies at the transition between the Sichuan Basin and surrounding low-mountain terrain, with elevations of 180–885m above sea level. The subtropical monsoon climate sup-

ports a mixed agricultural–forest landscape where paddy fields, dry cropland, orchards, and forest interleave at fine spatial scales. For double-anonymized review, the three experimental townships are reported as Township A, Township B, and Township C. The processing codes below preserve reproducibility of the experimental pipeline without disclosing owner attributes or original parcel geometries.

Three townships were selected to represent a gradient of spatial complexity:

- **Township A** (code 500227109): 3,607 parcels, 1,601 swappable (1,130 farmland + 471 forest), aggregated into 78 blocks.
- **Township B** (code 500227108): 5,500 parcels, 2,669 swappable (1,998 farmland + 671 forest), aggregated into 132 blocks.
- **Township C** (code 500227105): 12,950 parcels, 6,787 swappable (5,032 farmland + 1,755 forest), aggregated into 338 blocks.

Table 2: Characteristics of the three study townships and their block-level aggregation. “Init. *baimu*” indicates the number of *baimu fang* (≥ 6.67 ha contiguous farmland) present in the initial land-use configuration before any optimization.

Township	Total	Swappable	Farmland	Forest	Blocks	Parcels/block	Median area (ha)	Init. <i>baimu</i>
A (Small)	3,607	1,601	1,130	471	78	20	33	9
B (Medium)	5,500	2,669	1,998	671	132	20	28	6
C (Large)	12,950	6,787	5,032	1,755	338	20	20	6

3.2. Data sources

Cadastral data. Land-use parcel data were obtained from the Third National Land Survey (TNLS) database for Bishan District, completed in 2019, containing 101,657 polygon parcels with DLBM (*dilei bianma*, land-use classification) codes and parcel boundaries. The data are governed by the access regulations of China’s relevant land-administration authorities and are not redistributed in this paper; we provide synthetic test fixtures and an aggregate-statistics summary in the replication repository so that the methodology can be validated end-to-end without access to the original dataset (Section: Data and code availability).

Parcels were classified into three categories using the first three digits of the DLBM code: **farmland** (DLBM 011 paddy field, 012 irrigated dry land,

013 dry land), **forest** (031 arboreal forest, 032 shrub forest, 033 other forest land), and **other** (all remaining classes including 04x grassland, 06x urban / built-up land, 1xx transportation, 11x water, etc.). Only the farmland and forest classes are considered swappable; “other” parcels are treated as fixed barriers throughout the optimization. Geometric cleaning consisted of removing zero-area artifacts, repairing self-intersections via shapely `buffer(0)`, and discarding parcels with no DLBM code; no further smoothing was applied.

Coordinate reference system and area computation. All polygons were re-projected from the source CRS to EPSG:32648 (UTM zone 48N), in which units are metres and parcel areas are computed as planar polygon areas $a_i = \int_{\partial p_i} dA$ via shapely. The *baimu fang* threshold $A_{\text{baimu}} = 66,700 \text{ m}^2$ is consistent with this metric definition ($1 \text{ mu} = 666.7 \text{ m}^2$; $100 \text{ mu} = 66,700 \text{ m}^2$).

Slope. Slope was derived from the publicly available Copernicus DEM GLO-30 (30 m posting; Copernicus 5) using a planar slope operator on the DSM, expressed in degrees. A per-parcel mean slope s_i was assigned by zonal statistics: each parcel polygon was rasterized at the DEM resolution and s_i was computed as the unweighted mean of all DEM cells whose centre falls inside the polygon. Parcels smaller than a single DEM cell—a small fraction of the full database—were assigned the slope of the DEM cell containing the parcel centroid. No interpolation across parcel boundaries was applied.

Adjacency. Queen contiguity adjacency was constructed on the swappable subset using `libpysal`; two parcels are neighbours if their polygon boundaries share *any* point (vertex or edge). Construction-land, water, and transportation parcels are excluded from this graph and therefore act as physical barriers (Section 4.2). Additional features (elevation, aspect sine/cosine, shape index) were computed in the same projection and used in the per-block feature vector (Section 4.3).

Compliance. Use of the cadastral data is restricted to research; aggregate statistics, derived metrics, and code are released; no individual cadastral identifiers, owner attributes, or original geometries are released.

4. Methodology

4.1. Problem formulation

The farmland consolidation optimization is formulated as a tri-objective combinatorial problem. Given a set of parcels $\mathcal{P} = \{p_1, \dots, p_M\}$ with land-use types $l_i \in \{\text{farmland}, \text{forest}, \text{other}\}$, slope values s_i , and areas a_i , the goal is to find a land-use assignment $\mathbf{I}^*$ that simultaneously:

$$\text{Minimize: } \bar{S} = \frac{\sum_{i:l_i=F} s_i \cdot a_i}{\sum_{i:l_i=F} a_i} \quad (\text{area-weighted mean farmland slope}) \quad (1)$$

$$\text{Maximize: } \bar{C} = \frac{1}{|\mathcal{P}_F|} \sum_{i \in \mathcal{P}_F} \frac{|\{j \in \mathcal{N}(i) : l_j = F\}|}{|\mathcal{N}(i)|} \quad (\text{farmland contiguity}) \quad (2)$$

$$\text{Maximize: } B = \sum_{k=1}^K \mathbb{I}[\text{area}(\mathcal{C}_k) \geq A_{\text{baimu}}] \quad (\text{baimu fang count}) \quad (3)$$

where $\mathcal{N}(i)$ denotes the set of parcels adjacent to p_i (Queen contiguity), $\mathcal{C}_1, \dots, \mathcal{C}_K$ are the connected components of the farmland adjacency subgraph, and $A_{\text{baimu}} = 66,700 \text{ m}^2$ ($6.67 \text{ ha} = 100 \text{ mu}$) is the policy-defined threshold for a consolidated field. The optimization is subject to farmland count conservation ($|\mathcal{P}'_F| = |\mathcal{P}_F|$) and an exchange budget ($B_{\text{max}} = 100$ paired swaps).

The third objective (Eq. 3) is a contribution of this paper: *baimu fang* count is a discrete, non-differentiable, globally computed metric that depends on the connected component structure of the entire farmland subgraph. A single parcel conversion can merge two sub-threshold farmland clusters into a super-threshold consolidated field, creating a discontinuous reward signal that rewards strategic placement of exchanges.

4.2. Block-level spatial abstraction

The key methodological innovation is the aggregation of individual parcels into *consolidation blocks*—spatially coherent groups of farmland and forest parcels that serve as the decision units for the DRL agent. This aggregation serves three purposes: (1) it reduces the action space by approximately $20\times$ (from thousands of parcels to tens or hundreds of blocks), making training tractable; (2) it creates resource-competition dynamics where selecting one block consumes budget that could be invested elsewhere; and (3) it aligns the

agent’s decision granularity with the planning scale at which consolidation projects are actually designed and funded.

4.2.1. Hybrid barrier segmentation and clustering

Block construction proceeds in five steps:

Step 1: Identify swappable parcels. From the full parcel set, extract all farmland and forest parcels as swappable candidates. Non-swappable parcels (roads, waterways, construction land, etc.) are retained as spatial barriers.

Step 2: Build swappable-only adjacency. Construct a Queen contiguity graph on swappable parcels only, using the `libpysal` library. Because barrier parcels (roads, waterways) are excluded from the graph, they naturally segment the swappable parcels into disconnected clusters—parcels separated by a road or stream will not be adjacent in this graph, even if they are physically proximate.

Step 3: Connected component decomposition. Apply breadth-first search to identify connected components of the swappable adjacency graph. Each component represents a group of parcels enclosed by natural or infrastructure barriers. Many components are small (3–10 parcels), but some are very large (hundreds of parcels) where barriers are sparse.

Step 4: Constrained agglomerative clustering. Large components (those exceeding 30 parcels) are subdivided using Ward-linkage agglomerative clustering with a connectivity constraint, implemented via `sklearn.cluster.AgglomerativeClustering`. The connectivity matrix ensures that only spatially adjacent parcels can be merged, producing geographically contiguous sub-blocks. The target is approximately 20 parcels per block, chosen to balance the desire for fine-grained control (smaller blocks) with the need for sufficient within-block diversity to enable meaningful swaps (each block should contain both farmland and forest parcels).

Step 5: Filtering. Remove blocks with fewer than 3 parcels or less than 0.5 ha total area, as these are too small to support meaningful consolidation activity. The remaining parcels are assigned to their respective blocks.

This hybrid approach combines the physical interpretability of barrier-based segmentation (blocks respect real landscape boundaries) with the statistical control of clustering (blocks are approximately equal in size). Table 3 summarizes the block construction results. Fig. 2 illustrates the five-step block construction process on a representative area.

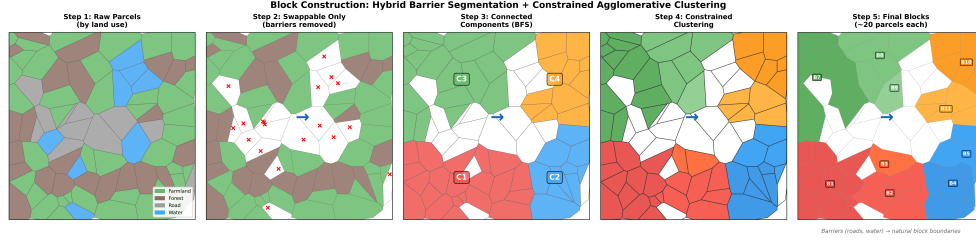


Figure 2: Block construction process illustrated on synthetic parcels resembling Township A. Step 1: raw parcels colored by land use (farmland=green, forest=brown, road=gray, water=blue). Step 2: barriers removed, only swappable parcels retained. Step 3: connected components identified via BFS. Step 4: large components subdivided by constrained Ward clustering. Step 5: final blocks labeled (~ 20 parcels each).

Table 3: Block construction statistics. “Both F+F” indicates the percentage of blocks containing both farmland and forest parcels (eligible for within-block swaps). “Assigned” indicates the percentage of swappable parcels successfully assigned to blocks.

Township	Blocks	Assigned (%)	Both F+F (%)	Median parcels/block	Median area (ha)
A (78 blocks)	78	98	99	20	33
B (132 blocks)	132	99	91	20	28
C (338 blocks)	338	98	97	20	20

4.3. Block-level Markov Decision Process

The optimization is cast as a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, \mathcal{T}, R, H)$ where the agent selects blocks to invest consolidation resources in.

4.3.1. State space

At step t , the state $\mathbf{s}_t$ consists of two components:

Per-block features $\mathbf{X}_t \in \mathbb{R}^{N_b \times K_B}$, where N_b is the number of blocks and $K_B = 17$ features per block:

- 1–4. Normalized mean slope, slope standard deviation, minimum slope, and maximum slope of all parcels within the block.
- 5–7. Block area (normalized), farmland count, and forest count within the block.
- 8–9. Farmland ratio and forest ratio within the block.
- 10–11. Mean farmland slope and mean forest slope within the block.
12. Contiguity index: mean proportion of same-type neighbors for farmland parcels within the block.

- 13. Compactness: isoperimetric quotient $4\pi A/P^2$ of the block’s convex hull.
- 14–15. Number of adjacent blocks and mean slope of adjacent blocks.
- 16. Budget remaining (normalized to $[0, 1]$).
- 17. Step progress (t/H).

Global state vector $\mathbf{g}_t \in \mathbb{R}^{K_G}$, where $K_G = 9$:

- 1–2. Normalized global mean farmland slope and mean forest slope.
- 3. Global farmland contiguity index.
- 4. Farmland count deviation from initial.
- 5. Budget utilization ratio.
- 6. Step progress.
- 7. *Baimu fang* count (current).
- 8. Total *baimu fang* area (normalized).
- 9. *Baimu fang* count change from initial.

The state dimension is $N_b \times 17 + 9$, which for the three townships evaluates to $78 \times 17 + 9 = 1,335$ (Township A), 2,253 (B), and 5,755 (C). Although these dimensions are large, the BlockScoringPolicy architecture (Section 4.5) processes each block independently through a shared scorer, so the number of *learnable parameters* is independent of N_b .

4.3.2. Action space

At each step, the agent selects one block $a_t \in \{1, 2, \dots, N_b\}$. An action mask $\mathbf{m}_t \in \{0, 1\}^{N_b}$ ensures that only blocks containing both *remaining unused* farmland and forest parcels are available:

$$m_t(b) = \mathbb{1} \left[n_F^{(b, \text{unused})}(t) > 0 \wedge n_L^{(b, \text{unused})}(t) > 0 \right] \quad (4)$$

where $n_F^{(b, \text{unused})}(t)$ and $n_L^{(b, \text{unused})}(t)$ are the counts of farmland and forest parcels in block b that have *not yet been involved in any swap* during the current episode. To enforce this, the environment maintains a per-parcel boolean mask $\mathbf{swapped}_i$ that is set to **true** as soon as parcel i participates in a paired exchange, and the within-block greedy engine (Section 4.3.3) draws candidates only from parcels with $\mathbf{swapped}_i = \mathbf{false}$. Each parcel can therefore be involved in at most one farmland–forest swap per episode, preventing repeated swapping or undoing of earlier decisions, and ensuring

that the budget $B_{\max}$ corresponds to a fixed number of distinct parcel-level changes rather than a count of potentially redundant operations. When either the unused-farmland or unused-forest count in a block reaches zero, the block is masked out for the rest of the episode.

4.3.3. Transition function: connectivity-aware greedy engine

Upon selecting block b , the environment executes κ farmland–forest swaps within that block ($\kappa = 5$ by default, consuming 5 units of the total budget $B_{\max} = 100$). Each swap proceeds as follows:

Farmland removal scoring. For each farmland parcel i in block b :

$$\text{score}_{\text{remove}}(i) = s_i - \delta \cdot |\{j \in \mathcal{N}(i) : l_j = F\}| \quad (5)$$

where $\delta = 0.5$ is the connectivity penalty. This score preferentially selects farmland parcels that are both steep (high s_i) and spatially isolated from other farmland (low farmland neighbor count), preserving the contiguity of the remaining farmland cluster.

Forest conversion scoring. For each forest parcel j in block b :

$$\text{score}_{\text{convert}}(j) = s_j - \gamma \cdot |\{k \in \mathcal{N}(j) : l_k = F\}| \quad (6)$$

where $\gamma = 1.0$ is the connectivity bonus (note the sign: lower score is selected, so high farmland neighbor count reduces the score, making farmland-adjacent forest parcels *preferred*). This converts forest parcels that are both gentle-slope (low s_j) and adjacent to existing farmland, directly promoting farmland contiguity and *baimu fang* formation.

Units of δ and γ . Equations 5 and 6 subtract a dimensionless integer (the same-type neighbour count) from a slope value in degrees. The penalty/bonus magnitudes therefore have units of degrees-per-neighbour: $\gamma = 1.0^\circ/\text{neighbour}$ means that one extra adjacent farmland parcel offsets one degree of slope in the forest-conversion ranking, while $\delta = 0.5^\circ/\text{neighbour}$ offsets one degree for the farmland-removal ranking. The slopes in the Bishan dataset span roughly $0\text{--}35^\circ$ and the same-type neighbour counts range $0\text{--}6$ in typical Queen contiguity, so the connectivity term and the slope term are of comparable magnitude in the score, and neither dominates the other. We treat (γ, δ) as fixed engine constants throughout the experiments and discuss their sensitivity in Section 6.4.

The asymmetry between $\delta = 0.5$ (removal penalty) and $\gamma = 1.0$ (conversion bonus) reflects a deliberate “active consolidation” bias: weighting the

reward for converting farmland-adjacent forest *twice as heavily* as the penalty for removing well-connected farmland. Intuitively, δ slows down the destruction of existing contiguous farmland clusters, while γ aggressively pulls in forest parcels at the boundaries of those clusters to enlarge them. The 2:1 ratio shifts the within-block engine from a passive “protect current edges” regime toward a “fill the gaps and grow the cluster” regime—the very behavior that pushes near-threshold farmland clusters past the 6.67 ha line and creates new *baimu fang*. The asymmetry is held fixed across all townships and all baselines so that the comparison among methods is internally consistent; its sensitivity is discussed in Section 6.4.

The farmland parcel with the highest $\text{score}_{\text{remove}}$ and the forest parcel with the lowest $\text{score}_{\text{convert}}$ are swapped, provided that the removed farmland’s slope exceeds the converted forest’s slope (otherwise the swap would worsen the mean farmland slope, and it is skipped). This process repeats κ times or until no improving swaps remain.

Key design property: spatial spillover. Because the *baimu fang* metric and the contiguity index are computed over the *entire* township (crossing block boundaries), swaps within one block can change the connectivity-aware scores of parcels in *adjacent* blocks. For example, converting a forest parcel to farmland at the boundary of block b increases the farmland neighbor count of parcels in an adjacent block b' , making those parcels more attractive for future forest-to-farmland conversion. This cross-block spatial spillover creates observable sequential dependencies in the block-level formulation: the value of investing in block b' at step $t + 1$ depends on whether block b was selected at step t . This is the kind of inter-step dependency that the proposed abstraction is designed to expose to a planning policy.

4.3.4. Reward function

The reward at step t combines four components, each defined as a *normalized* improvement signal so that the four channels are dimensionally comparable:

$$r_t = w_S \cdot \widetilde{\Delta S}_t + w_C \cdot \widetilde{\Delta C}_t + w_B \cdot \widetilde{\Delta A}_t^B + b_B \cdot \Delta n_t^{B,+} \quad (7)$$

where, denoting the area-weighted mean farmland slope by $\bar{S}_t$, the contiguity index by $\bar{C}_t$, the total *baimu fang* area by A_t^B , the *baimu fang* count by n_t^B ,

and the values at $t=0$ by $\bar{S}_0, \bar{C}_0, A_0^F$ (initial total farmland area):

$$\widetilde{\Delta S}_t = \frac{\bar{S}_{t-1} - \bar{S}_t}{|\bar{S}_0| + \varepsilon}, \quad \widetilde{\Delta C}_t = \frac{\bar{C}_t - \bar{C}_{t-1}}{|\bar{C}_0| + \varepsilon}, \quad (8)$$

$$\widetilde{\Delta A}_t^B = \frac{A_t^B - A_{t-1}^B}{A_0^F + \varepsilon}, \quad \Delta n_t^{B,+} = \max(0, n_t^B - n_{t-1}^B). \quad (9)$$

With this convention, every channel produces a positive reward when the corresponding indicator improves: $\widetilde{\Delta S}_t > 0$ when slope decreases, $\widetilde{\Delta C}_t > 0$ when contiguity rises, $\widetilde{\Delta A}_t^B > 0$ when consolidated-field area expands, and $\Delta n_t^{B,+} > 0$ only when a *new baimu fang* is formed (existing fields lost are not penalized through this channel; their loss is reflected via the area term). The weights are $w_S = 2000$, $w_C = 500$, $w_B = 500$, $b_B = 20$, identical for all three townships and all 5 seeds (Table 5). They define one policy-preference scenario rather than a universal social welfare function. Table 4 gives the planning interpretation of each component and how a practitioner would change it under different local priorities.

Table 4: Policy interpretation of the reward components. The reported experiments use one fixed preference setting; alternative local priorities require re-running the scenario comparison rather than treating these weights as normative.

Reward term	Weight	Planning meaning in this study	If local policy priority differs
$\widetilde{\Delta S}_t$	$w_S = 2000$	Emphasizes reduction of mean farmland slope and erosion-related production risk	Increase when slope control dominates; decrease when field-block provision dominates
$\widetilde{\Delta C}_t$	$w_C = 500$	Rewards local farmland contiguity and reduced fragmentation	Increase when contiguous operation is the primary target
$\widetilde{\Delta A}_t^B$	$w_B = 500$	Rewards growth of total area inside qualifying field blocks	Increase when consolidated-field area is more important than slope reduction
$\Delta n_t^{B,+}$	$b_B = 20$	Gives a discrete bonus when a new threshold-qualified field appears	Increase when the number of project-recognized fields is a key assessment target

A penalty of -1.0 is applied when a selected block yields zero completed swaps (e.g., all available swaps would worsen slope), discouraging wasteful block selections. Because the four channels are normalized to $\sim O(1)$ per-step magnitudes, the per-step reward typically falls in $[-1, 1]$ and the cumulative episode return is on the order of tens to a few hundreds—consistent with the per-seed total rewards reported in Table 13 (Township A ≈ 150 , B ≈ 38 , C ≈ 266).

The connectivity-aware greedy engine (Section 4.3.3) guarantees that every farmland removal is paired with a forest conversion within the same step, so the parcel *count* of each land-use class is conserved exactly by construction. We discuss the related question of farmland *area* conservation—which the parcel-pairing rule does *not* guarantee—in Section 6.4 as a known limitation that informs the interpretation of the reported metrics.

4.3.5. Episode structure

Each episode consists of $H = 20$ steps. At each step, $\kappa = 5$ swaps are executed within the selected block, for a total budget of $H \times \kappa = 100$ paired swaps per episode. The choice of $\kappa = 5$ balances two competing considerations: (1) *sufficient within-block impact*—with fewer than 3–5 swaps per block visit, the greedy engine cannot substantially alter the block’s spatial configuration, weakening the reward signal for block selection; and (2) *sufficient macroscopic decisions*—the agent needs enough steps ($H = B_{\max}/\kappa$) to express a meaningful sequential strategy. At $\kappa = 1$, the agent makes 100 small block selections, which we use as a proxy for the near-independent regime rather than as a full parcel-level DRL benchmark; at $\kappa = 20$, only 5 macro-decisions remain, insufficient for the “invest-then-recover” temporal strategies observed on Township C (Section 5.4). The value $\kappa = 5$ ($H = 20$) provides a practical middle ground; an explicit ablation over $\kappa \in \{1, 3, 5, 10, 20\}$ is reported in Section 5.7 (Table 14). The discount factor is $\gamma_{\text{RL}} = 0.99$, corresponding to an effective planning horizon of ~ 100 steps (well beyond the 20-step episode), ensuring that the agent considers the full-episode consequences of early block selections.

4.4. Baimu fang computation

Baimu fang identification is performed via breadth-first search (BFS) on the farmland adjacency subgraph after each step. The algorithm operates as follows:

1. Construct the subgraph of all farmland parcels using the pre-computed Queen contiguity adjacency.
2. Identify connected components via BFS.
3. For each component, sum the areas of all constituent parcels.
4. A component qualifies as a *baimu fang* if its total area $\geq A_{\text{baimu}} = 66,700 \text{ m}^2$ (6.67 ha).
5. Return the count and total area of qualifying components.

This computation operates on the *full township* farmland graph, not restricted to individual blocks. Thus, a *baimu fang* can span multiple blocks, and swaps in one block can contribute to forming a consolidated field in a neighboring block—reinforcing the spatial spillover mechanism described in Section 4.3.3.

The BFS computation requires approximately 1.3ms per call on the largest township (12,950 parcels), adding negligible overhead to training (which involves $\sim 200,000$ such calls).

4.5. Policy network architecture

The policy network, which we call *BlockScoringPolicy*, follows a shared-scorer design. It evaluates each block independently with the same neural scorer, so the number of *learnable parameters* is independent of the number of blocks N_b . This design is defined here as part of the block-level method and can be reimplemented from the equations below. The architecture consists of two sub-networks:

Scorer network. For each block $b \in \{1, \dots, N_b\}$, the per-block feature vector $\mathbf{x}_b \in \mathbb{R}^{17}$ (the 17 local features enumerated in Section 4.3, columns of $\mathbf{X}_t$) is concatenated with the global feature vector $\mathbf{g}_t \in \mathbb{R}^9$ to form a 26-dimensional input $[\mathbf{x}_b; \mathbf{g}_t] \in \mathbb{R}^{26}$. This input is mapped to a scalar logit:

$$\text{logit}_b = \text{Scorer}([\mathbf{x}_b; \mathbf{g}_t]) \in \mathbb{R}, \quad \text{Scorer} : \mathbb{R}^{26} \rightarrow \mathbb{R} \quad (10)$$

implemented as a two-layer multilayer perceptron (MLP) with structure $26 \rightarrow 128 \rightarrow 64 \rightarrow 1$ and tanh activations. *Crucially, the same Scorer weights are applied to every block:* the network is invoked N_b times per state, once per block, with the global vector $\mathbf{g}_t$ shared across invocations and only $\mathbf{x}_b$ varying. The N_b resulting logits are then assembled into a vector and passed through a masked softmax that zeroes out blocks ineligible under the action mask (Section 4.3):

$$\pi(a_t = b \mid \mathbf{s}_t) = \frac{\exp(\text{logit}_b) \mathbb{I}[m_t(b) = 1]}{\sum_{b'=1}^{N_b} \exp(\text{logit}_{b'}) \mathbb{I}[m_t(b') = 1]}. \quad (11)$$

This dimension-invariant construction means a single trained set of weights can in principle be deployed across townships with different N_b values, although we do not exploit this transfer capability in the present study (each township is trained independently).

Value network. The state value depends only on global features:

$$V(\mathbf{s}_t) = \text{Value}(\mathbf{g}_t) \in \mathbb{R}, \quad \text{Value} : \mathbb{R}^9 \rightarrow \mathbb{R} \quad (12)$$

implemented as $9 \rightarrow 64 \rightarrow 32 \rightarrow 1$ with tanh activations. Restricting the critic to global features keeps the value-function input dimension constant

across townships (matching the policy’s dimension invariance) and avoids the need for any pooling over a variable number of blocks.

Parameter count and inference cost. The total parameter count is 14,530 (Scorer: $26 \times 128 + 128 + 128 \times 64 + 64 + 64 \times 1 + 1 = 11,713$; Value: $9 \times 64 + 64 + 64 \times 32 + 32 + 32 \times 1 + 1 = 2,721$; with bias terms). This count is identical regardless of N_b , which is why the architecture scales gracefully from 78 blocks (Township A) to 338 blocks (Township C). Per-state inference, dominated by N_b Scorer evaluations, is well under 1 ms even on the largest township, making the framework cheap at deployment time and avoiding the overhead of graph neural networks that would otherwise be tempting given the spatial structure of the problem.

Weights are initialized using orthogonal initialization [15] with gain 0.01 for the scorer (encouraging near-uniform initial action probabilities) and gain 1.0 for the value network.

4.6. Training

All models are trained using Maskable PPO from the `sb3-contrib` library [9]. Table 5 lists the hyperparameters.

Table 5: Training hyperparameters for the block-level MDP.

Hyperparameter	Value
Algorithm	Maskable PPO
Learning rate	1×10^{-3} (Adam)
Rollout length (n_{steps})	512
Minibatch size	256
PPO epochs	10
Discount factor (γ_{RL})	0.99
GAE parameter (λ)	0.95
Clip range	0.2
Entropy coefficient	0.01
Total timesteps	200,000
Episode length (H)	20 steps
Swaps per step (κ)	5
Total budget (B_{max})	100 paired swaps

Why 200,000 timesteps suffice. The 20-step episode length means that 200,000 timesteps correspond to 10,000 episodes per seed—a comfort-

able margin above the empirically observed plateau, which on all three townships occurs at $\approx 50,000$ timesteps ($\sim 2,500$ episodes; Fig. 3). The remaining $\sim 150,000$ timesteps serve as a buffer to confirm that the plateau is stable rather than transient. The block-level formulation’s relatively small action space (78–338 blocks) keeps each PPO update inexpensive even on the largest township.

For each of the three townships, 5 models are trained with random seeds 0–4, yielding 15 trained models total. Training was conducted on NVIDIA A100 40GB GPUs, with per-township training times of approximately 25 minutes (A), 50 minutes (B), and 2 hours (C).

4.7. Baseline methods

We compare the block-level DRL agent against five baselines that operate in the same block-level framework (20 steps, $\kappa = 5$ swaps per step, total budget 100). *All baselines share the same connectivity-aware greedy engine for within-block execution*; they differ *only* in their block-selection rule. Four of the five baselines do not see the multi-objective reward; the fifth (Reward-Greedy) does, and is included specifically to isolate how much of any DRL advantage is attributable to multi-step planning rather than to access to the multi-objective signal.

Greedy-Global. At each step, evaluates all N_b blocks and selects the one whose greedy execution yields the greatest single-step *slope* improvement. This is the strongest single-objective slope baseline at the block level, conceptually analogous to the slope-only Greedy baselines that have long been used in farmland consolidation heuristics.

Greedy-Sequential. Iterates through blocks in a fixed order (sorted by initial mean farmland slope, descending). Each block receives one step of investment before moving to the next; after cycling through all blocks, the sequence repeats.

Random-Block. At each step, selects a block uniformly at random from those with valid action masks. Averaged over 5 random seeds.

Round-Robin. Visits all blocks in sequence, allocating one step of investment to each. Blocks with more farmland and forest receive proportionally more attention over the episode.

Reward-Greedy (one-step lookahead, multi-objective). At each step, evaluates all N_b unmasked blocks by simulating the within-block greedy engine on a copy of the environment state, computing the same multi-component reward Eq. 7 that DRL is trained on (with the same weights

$w_S = 2000$, $w_C = 500$, $w_B = 500$, $b_B = 20$), and selecting the block with the largest one-step reward. This is a strong reward-aware myopic baseline and a planning-horizon ablation of DRL: any performance gap between Reward-Greedy and DRL is attributable to the multi-step planning and value-function bootstrapping that the RL agent provides on top of one-step reward maximization.

For Township B, where the main DRL-specific advantage appears, we also run two short-horizon reward-aware search baselines as robustness checks. The depth-2 lookahead enumerates all valid two-step block sequences from the current state at each decision point and selects the first block of the sequence with the largest cumulative simulated reward. The depth-3 lookahead uses the same recursive simulation but keeps only the top eight partial sequences after each expansion, which makes the search tractable on the 132-block action space. Both short-horizon baselines use the same reward weights, action masks, 20-step episode length, and within-block execution engine as DRL and Reward-Greedy. They are reported as Township-B stress checks rather than as full cross-township baselines, because exhaustive lookahead cost grows quickly with the number of blocks and with search depth.

All baselines use identical within-block execution (the connectivity-aware greedy engine with $\gamma = 1.0$, $\delta = 0.5$).

4.8. Evaluation

Trained models are evaluated with deterministic inference: at each step, the trained policy selects the block with the highest logit score (argmax over unmasked blocks). Each evaluation produces a single deterministic trajectory. We report slope change (%), contiguity change, *baimu fang* count change, *baimu fang* area change (ha), and budget utilization across all 5 seeds. Because the current swap engine conserves parcel count rather than total farmland area, these metrics are interpreted as scenario-screening indicators rather than legally area-balanced implementation outcomes.

We therefore add two post-hoc policy-risk checks. First, an area-drift audit replays every final configuration and reports the net change in total farmland area. Second, a Township-B area-tolerance stress test modifies the transition engine so that a candidate paired swap is accepted only if the cumulative farmland-area change remains within a specified tolerance and the exact area-weighted mean farmland slope improves. The stress test is run at 0.5% and 1.0% cumulative farmland-area tolerances. The deterministic baselines are rerun under this modified engine; for DRL, we replay the

already learned block histories rather than retraining an area-constrained policy. The DRL rows in this stress test should therefore be read as history-replay diagnostics, not as evidence that a constrained DRL agent has been optimized.

5. Results

5.1. Training convergence

Fig. 3 shows the training dynamics for all three townships (A, B, C). The episodic reward exhibits clear learning progress, with reward curves flattening at approximately 50,000 timesteps ($\sim 2,500$ episodes) in all cases. We use “convergence” in an operational training-stability sense: a seed is treated as converged when the reward curve reaches a stable plateau and the deterministic evaluation produces a non-degenerate trajectory with positive slope reduction. This is not a claim of global optimality or Pareto-front convergence.

Table 6 reports the convergence outcome of the block-level approach across all 15 seed–township combinations.

Table 6: Operational training convergence on the three Bishan townships under the tested settings (5 seeds per township, identical hyperparameters, 200,000 timesteps). A seed is counted as converged when the training reward has reached a stable plateau and deterministic evaluation produces a non-degenerate trajectory with positive slope reduction relative to the initial configuration. This table reports training stability, not proof of global optimality.

Township	Block-level convergence
A (Small, 3,607 parcels, 78 blocks)	5/5 (100%)
B (Medium, 5,500 parcels, 132 blocks)	5/5 (100%)
C (Large, 12,950 parcels, 338 blocks)	5/5 (100%)
Total	15/15 (100%)

Training stability is uniformly tight: on Township C the across-seed reward coefficient of variation is $<1\%$ (Table 13), and no seed–township combination produced a degenerate or bimodal trajectory. Training reward curves on all three townships (Fig. 3) cluster tightly across seeds and plateau at approximately 50,000 timesteps ($\sim 2,500$ episodes), well within the 200,000-timestep budget. The remainder of this section unpacks the multi-objective behaviour of the stabilized policies.

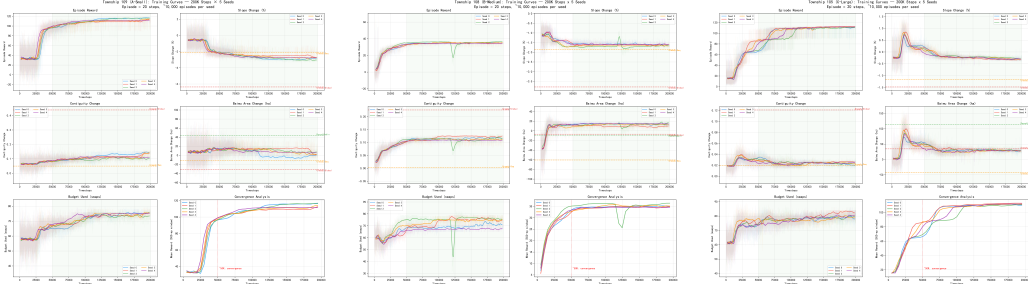


Figure 3: Training curves for Township A (left, 78 blocks), Township B (middle, 132 blocks), and Township C (right, 338 blocks), showing 5 seeds each with 200-episode moving average. The reward and evaluation indicators stabilize by $\sim 50K$ timesteps. The seed trajectories cluster tightly throughout training, with no seed in any township producing a degenerate or bimodal trajectory under the tested settings.

5.2. Cross-township comparison

Table 7 presents the comprehensive results across all methods and townships.

Several patterns emerge:

DRL is the only main-table method with positive *baimu fang* area on Township B. On Township B every other cross-township method, including the strong reward-aware one-step lookahead Reward-Greedy, produces negative *baimu fang* area (ranging from -6.9 ha for Greedy-Global to -101.9 ha for Reward-Greedy). Only DRL achieves a positive value ($+12.9 \pm 3.2$ ha). Crucially, Reward-Greedy on Township B has access to the *same* multi-objective reward DRL optimizes (with the same weights) and yet produces a severe *baimu fang* area loss, because the slope channel (whose magnitude scales with $w_S = 2000$) dominates the per-step decision and pushes the engine to remove high-area qualifying farmland whenever doing so reduces the area-weighted slope mean. DRL avoids this trap under the tested protocol, indicating that the gain on this objective comes from multi-step planning rather than from access to the multi-objective signal alone.

Reward-Greedy is competitive with, and on some metrics surpasses, DRL on Townships A and C. On Township A, Reward-Greedy achieves slightly better slope (-3.03% vs. DRL -2.61%), better contiguity ($+0.198$ vs. $+0.130$), the same *baimu fang* count change (0, joint best), and slightly better *baimu fang* area ($+13.5$ ha vs. $+8.8$ ha). On Township C, Reward-Greedy achieves better slope (-0.52% vs. -0.33%), comparable

Table 7: Cross-township comparison of all methods. DRL results report mean $\pm$ std across 5 seeds. Slope change: more negative is better; contiguity, *baimu fang* count, and *baimu fang* area changes: more positive is better. Best result per metric per township in bold; ties on the *baimu fang* count column are bolded jointly. Reward-Greedy is a one-step lookahead baseline that, at every step, simulates each unmasked block under the same engine and picks the block whose one-step reward (Eq. 7) is largest. It is the strongest *multi-objective* baseline and the natural ablation that isolates how much of DRL’s advantage comes from *multi-step* planning rather than from access to the multi-objective signal alone.

Township	Method	Slope (%)		Cont.	Baimu cnt	Baimu ha
A (78 blk)	Greedy-Global	−6.36		+0.439	−2	−31.5
	Greedy-Seq	−2.12		+0.050	−1	−12.0
	Random (avg)	−0.79		+0.051	0	+1.4
	Round-Robin	−0.33		+0.087	−1	+43.8
	Reward-Greedy	−3.03		+0.198	0	+13.5
	DRL	−2.61 $\pm$ 0.44	+0.130 $\pm$ 0.025		0	+8.8 $\pm$ 10.7
B (132 blk)	Greedy-Global	−3.79		+0.226	−1	−6.9
	Greedy-Seq	−1.85		+0.003	0	−60.4
	Random (avg)	−1.31		+0.032	−0.6	−38.1
	Round-Robin	−0.28		+0.062	−1	−8.9
	Reward-Greedy	−4.24		+0.036	0	−101.9
	DRL	−1.64 $\pm$ 0.04	+0.115 $\pm$ 0.009		−1	+12.9 $\pm$ 3.2
C (338 blk)	Greedy-Global	−1.48		+0.121	−1	+34.6
	Greedy-Seq	−1.17		+0.020	0	−43.0
	Random (avg)	−0.24		+0.026	+0.8	+9.6
	Round-Robin	+1.10		+0.047	+2	+114.4
	Reward-Greedy	−0.52		+0.045	+6	+28.0
	DRL	−0.33 $\pm$ 0.01	+0.023 $\pm$ 0.003		+5 $\pm$ 0	+27.7 $\pm$ 1.3

contiguity, a higher *baimu fang* count change (+6 vs. +5), and similar *baimu fang* area (+28.0 ha vs. +27.7 ha). On these two townships a one-step reward-aware lookahead does most of what the trained DRL agent does. This is an important and honest finding: *block-level abstraction is doing most of the work in this paper, not RL-induced multi-step planning per se*. The remaining DRL-specific gain shows up only on Township B, where Reward-Greedy’s myopic preference for the slope channel sacrifices consolidated-field area in a way DRL learns to avoid. We discuss the implication—that the right framing of the contribution is “block-level abstraction with a learnable scheduler,” not “RL beats heuristics across the board”—in Sections 6 and 6.4.

Greedy-Global is the strongest single-objective slope baseline; Reward-Greedy is the strongest cross-township multi-objective baseline. Greedy-Global achieves the best slope reduction on Townships A and C (−6.36%, −1.48%), and the best contiguity on A and C (+0.439, +0.121); but it produces negative *baimu fang* count change on all three townships and negative *baimu fang* area on Townships A and B. Reward-Greedy improves on Greedy-Global on A and C across all four metrics simultaneously by trading some slope for substantial *baimu fang* count and area gains. On Township B, Reward-Greedy attains the best slope reduction among the cross-township methods (−4.24%) but at the cost of severe *baimu fang* area loss, a clear illustration of the multi-objective tension that motivates the use of DRL on this township.

The Township-B advantage is not removed by stronger short-horizon search. Table 8 compares DRL with Reward-Greedy and two additional reward-aware lookahead baselines on the same township. Depth-2 exhaustive lookahead and depth-3 beam search both closely match Reward-Greedy on slope reduction (−4.23% and −4.26%) but still collapse *baimu fang* area (−108.8 ha and −103.8 ha). This result narrows the interpretation of DRL’s advantage: it is not merely that Reward-Greedy looks one step ahead. Under the tested reward weights and Township-B geometry, short-horizon reward search remains dominated by the slope channel, while the learned policy preserves more consolidated-field area at the cost of weaker slope reduction.

Fig. 4 visualizes the cross-township comparison across all four metrics, highlighting DRL’s distinctive multi-objective profile.

5.3. Policy-risk checks: area balance and engineering shape

The original engine conserved farmland parcel count but not farmland area, so we audited the final area balance before interpreting the scenarios for

Table 8: Township B policy-robustness checks against stronger short-horizon planning baselines. Depth-2 lookahead evaluates all valid two-step block sequences at each decision point; depth-3 lookahead uses beam width 8 for tractability. Both lookahead baselines use the same within-block execution engine and reward weights as DRL and Reward-Greedy. Slope change is better when more negative; contiguity, *baimu fang* count, and *baimu fang* area are better when more positive.

Method	Slope (%)	Cont.	<i>Baimu</i> cnt	<i>Baimu</i> ha
DRL mean	−1.64	+0.115	−1	+12.9
Reward-Greedy	−4.24	+0.036	+0	−101.9
Lookahead d=2, all actions	−4.23	+0.013	+1	−108.8
Lookahead d=3, beam 8	−4.26	+0.035	+0	−103.8

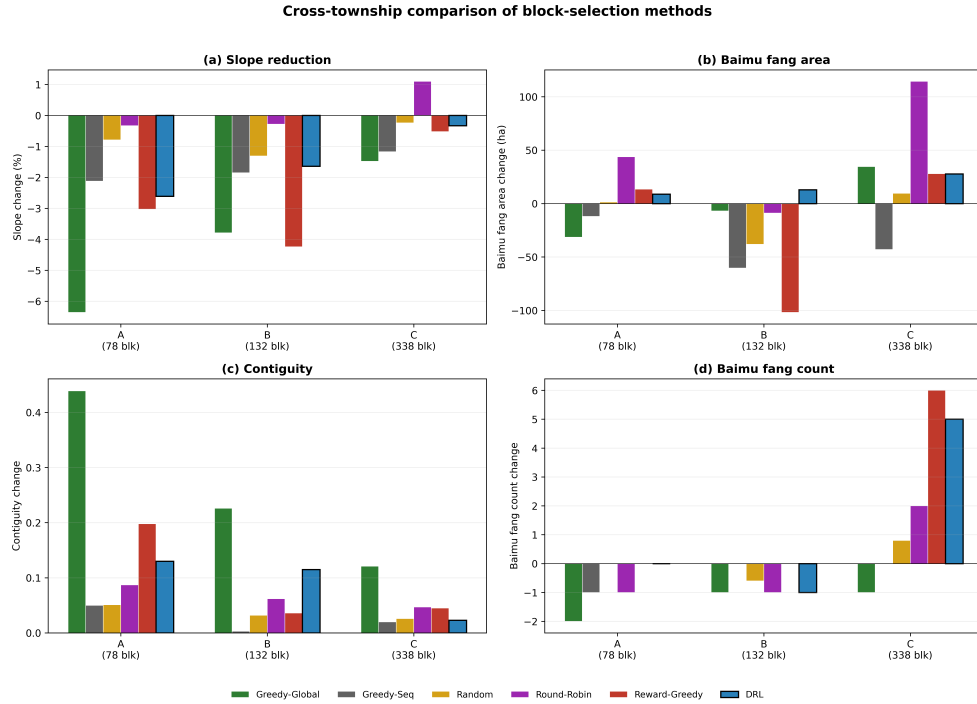


Figure 4: Cross-township comparison of all methods across four metrics: (a) slope reduction, (b) *baimu fang* area change, (c) contiguity change, and (d) new *baimu fang* count. DRL bars are outlined in black. DRL is uniquely the only main-table method with positive *baimu fang* area on Township B, even though its *baimu fang* count change on that township is negative.

Table 9: Policy-facing interpretation of the cross-township results. The table translates the optimization outputs into screening implications and implementation risks rather than treating any method as an automatic plan generator.

Township	Main trade-off observed	Screening implication	Boundary before implementation
A	Slope-oriented and reward-aware rules are both viable; DRL is balanced but not best on most metrics	A simple reward-aware greedy rule may be sufficient for first-pass scenario comparison	Confirm area balance and field compactness before treating any retained <i>baimu fang</i> as project-ready
B	Slope reduction conflicts strongly with retaining qualifying field-block area; greedy reward search sacrifices <i>baimu fang</i> area	Learned scheduling is most useful where one-step slope gains dis-mantle high-area qualifying fields	Area-tolerant replay removes the positive <i>baimu fang</i> area outcome, so constrained retraining is required
C	New field-block count is easy to increase once blocks are evaluated with the reward signal; Reward-Greedy and DRL are both effective	Reward-aware block screening may be enough when the policy target is count formation rather than slope maximization	Round-Robin maximizes area but raises slope, so local priorities must determine the acceptable trade-off

policy use. Table 10 reports the net farmland-area change for every method and township. The DRL scenarios had small average area drift in all three townships: $+0.14 \pm 0.45\%$ on A, $+0.28 \pm 0.12\%$ on B, and $+0.13 \pm 0.03\%$ on C. The largest Township-B drift among the main deterministic comparators was Reward-Greedy (-3.00%), which is also the method that produced the largest *baimu fang* area loss.

Table 10: Farmland-area balance audit. The optimization engine conserves farmland parcel count, not farmland area. This post-hoc audit replays each final configuration and reports the net farmland-area change. Random-Block and DRL are reported as mean $\pm$ standard deviation over five seeds; deterministic baselines report a single trajectory. Values are derived from controlled-access parcel geometry and released here only as aggregate diagnostics.

Township	Method	Δ farmland area (ha)	Δ farmland area (%)
A	Greedy-Global	-43.6	-2.06
A	Greedy-Sequential	-12.6	-0.60
A	Random-Block	-2.4 ± 13.0	-0.11 ± 0.61
A	Round-Robin	+31.2	+1.47
A	Reward-Greedy	+4.0	+0.19
A	DRL	$+2.9 \pm 9.6$	$+0.14 \pm 0.45$
B	Greedy-Global	-13.7	-0.40
B	Greedy-Sequential	-59.1	-1.74
B	Random-Block	-37.8 ± 25.9	-1.12 ± 0.77
B	Round-Robin	-11.6	-0.34
B	Reward-Greedy	-101.6	-3.00
B	DRL	$+9.6 \pm 4.0$	$+0.28 \pm 0.12$
C	Greedy-Global	+23.2	+0.40
C	Greedy-Sequential	-38.5	-0.67
C	Random-Block	$+7.7 \pm 30.2$	$+0.13 \pm 0.52$
C	Round-Robin	+89.8	+1.56
C	Reward-Greedy	+7.2	+0.12
C	DRL	$+7.4 \pm 1.9$	$+0.13 \pm 0.03$

This audit reduces one concern but does not remove the implementation boundary. A small net drift after 100 swaps is not equivalent to enforcing farmland occupation–compensation balance at every step. We therefore ran a stricter Township-B transition check in which every accepted swap had to keep cumulative farmland-area change within a tolerance and had to improve

the exact area-weighted mean farmland slope (Table 11).

Table 11: Township B area-tolerance transition check. The modified engine enforces cumulative farmland-area tolerance and uses an exact area-weighted mean-slope improvement check for paired swaps. DRL rows replay existing block histories only and are stress tests, not retrained policies.

Tolerance (%)	Method	Slope (%)	Cont.	Baimu cnt	Baimu ha	Farm area (%)	Budget
0.5	Greedy-Global	-3.79	+0.226	-1.0	-6.9	-0.40	100.0
0.5	Greedy-Sequential	-1.26	+0.027	+0.0	-14.1	-0.42	64.0
0.5	Round-Robin	-0.39	+0.045	-1.0	-14.1	-0.44	38.0
0.5	Reward-Greedy	-2.52	+0.123	+1.0	-13.3	-0.45	85.0
0.5	DRL replay mean	-1.93	+0.091	+0.0	-9.8	-0.36	77.6
1.0	Greedy-Global	-3.79	+0.226	-1.0	-6.9	-0.40	100.0
1.0	Greedy-Sequential	-1.78	+0.014	+0.0	-33.5	-0.92	65.0
1.0	Round-Robin	-0.46	+0.045	-1.0	-20.2	-0.61	45.0
1.0	Reward-Greedy	-3.12	+0.089	+1.0	-34.7	-0.99	80.0
1.0	DRL replay mean	-2.59	+0.081	+0.0	-29.7	-0.87	77.6

The area-tolerance stress test changed the Township-B policy interpretation. At a 0.5% tolerance, replaying the existing DRL block histories produced slope reduction (-1.93%), contiguity improvement ($+0.091$), and near-balanced farmland area (-0.36%), but the *baimu fang* area outcome fell to -9.8 ha. At a 1.0% tolerance, DRL replay improved slope more strongly (-2.59%) but lost 29.7 ha of *baimu fang* area. These are not failures of a retrained constrained DRL agent, because no such agent was trained here. They show that the unconstrained DRL scenario should be treated as an opportunity signal that still requires area-balanced retraining or optimization before implementation.

The second policy-risk check concerns field shape. The *baimu fang* count reported above is topological: any Queen-connected farmland component above 6.67 ha qualifies. High-standard farmland standards also require practical field shape for mechanized operation, so we replayed every DRL evaluation episode, dissolved final farmland connected components, and computed the isoperimetric quotient $IPQ = 4\pi A/P^2$ under three thresholds (Table 12).

The compactness audit shows that raw *baimu fang* counts are upper-bound estimates. Initial components are already irregular under Queen contiguity, and no initial or DRL-final component reaches $IPQ \geq 0.20$ in any township. DRL improves topological consolidation in some cases, especially Township C, but compactness gains are mixed because compactness is not part of the reward. A project-ready version should therefore use a shape-aware *baimu fang* definition, such as compactness-filtered BFS, Rook adja-

Table 12: Post-hoc compactness audit of *baimu fang* formed by the block-level DRL agent. For each township and each evaluation seed we recover the final farmland configuration by replaying the agent’s block selection sequence, identify all connected farmland components with area ≥ 6.67 ha (the raw *baimu fang* count), and compute the isoperimetric quotient $IPQ = 4\pi A/P^2$ on the dissolved farmland polygon of each component ($IPQ=1$ for a circle; lower means more elongated/fragmented shape). The “Init. IPQ” columns report the same compactness counts on the *initial* land-use configuration before any optimization, providing a fairness baseline: the Queen-contiguity connectivity definition combined with real cadastral parcel shapes typically yields low-IPQ components even prior to consolidation, so the post-DRL counts at each τ should be read against the corresponding initial value rather than against an ideal of $\tau=1$. DRL values are means over 5 seeds.

Township	Initial state				After DRL optimization (mean over 5 seeds)				
	Init. raw	Init. IPQ ≥ 0.10	Init. IPQ ≥ 0.20	Init. IPQ ≥ 0.30	Final raw	Final IPQ ≥ 0.10	Final IPQ ≥ 0.20	Final IPQ ≥ 0.30	Δ raw
A	9	3	0	0	9.0	0.6	0.0	0.0	+0.0
B	6	1	0	0	5.0	1.6	0.0	0.0	-1.0
C	6	0	0	0	11.0	1.0	0.0	0.0	+5.0

cency, or boundary-length-weighted connectivity.

5.4. Emergent township-specific strategies

A striking finding is that the DRL agent, trained with *identical hyperparameters* across all three townships, develops qualitatively different optimization strategies adapted to each township’s spatial structure:

Township A (balanced strategy). The agent achieves moderate slope reduction (-2.61%), the third-strongest contiguity improvement of any method ($+0.130$), and preserves existing *baimu fang* while slightly growing their area ($+8.8$ ha). On this township the reward-aware Reward-Greedy baseline is slightly stronger than DRL on slope, contiguity, and *baimu fang* area (Section 5.2); we therefore do not claim that DRL Pareto-dominates here, but rather report a balanced multi-objective profile (no single objective is sacrificed, no objective is best either) that emerges naturally from training without any per-township weight tuning. The cross-seed std on slope is $\pm 0.44\%$, the largest of the three townships.

Township B (contiguity-and-*baimu fang* area strategy). Under the same reward weights used on Townships A and C, the agent on Township B produces a strategy that delivers strong contiguity improvement ($+0.115$) while being the *only* tested method achieving positive *baimu fang* area ($+12.9$ ha), including the reward-aware Reward-Greedy and short-horizon lookahead baselines. Slope reduction is moderate (-1.64%) compared to the slope-optimal Reward-Greedy (-4.24%) and Greedy-Global (-3.79%). Both

DRL and Reward-Greedy visit 20 distinct blocks over the 20-step episode; the difference is *which* 20 blocks they pick. Reward-Greedy’s first choices are the township’s highest-slope blocks, which on Township B coincide with existing consolidated farmland clusters, so each early slope gain can remove high-area qualifying farmland (Fig. 5, middle and bottom panels). DRL’s first choices instead include blocks that sit *inside* or on the gentle-slope side of existing clusters, where the within-block engine can pair a low-slope forest conversion with a moderately-sloped farmland removal while retaining more qualifying area. The cumulative *baimu fang* area trace ends at +10.5 ha (seed 0; the 5-seed mean is $+12.9 \pm 3.2$ ha), and the extremely low seed variance (slope std = 0.04%) indicates a highly stable strategy.

Township C (*baimu fang*-focused strategy). The agent achieves only modest slope reduction (−0.33%) but forms 5 new consolidated fields with +27.7 ha total area. Block trajectory analysis (Fig. 8) reveals a two-phase “invest-then-recover” pattern in which the first ten steps pay an early slope penalty (peaking at +0.93% at step 1) to push near-threshold farmland clusters across the 6.67 ha line, after which the count is held at +5 and the slope channel is recovered to −0.33%. We caution that the formation *gain* on this township is not exclusive to multi-step planning: the Reward-Greedy one-step lookahead baseline introduced in Section 4.7 forms +6 *baimu fang* on Township C without any RL training. The DRL trajectory still illustrates how the planning horizon shifts the agent’s objective mix step-by-step, but the count metric on this township is more a property of block-level multi-objective evaluation than of multi-step credit assignment.

The emergence of these distinct strategies from a single set of hyperparameters is a key advantage of the DRL framework: the agent automatically adapts its behavior to the local spatial structure and opportunity landscape, without requiring manual strategy design or per-township parameter tuning.

5.5. Per-seed consistency

Table 13 reports per-seed results for all three townships.

Township C exhibits the most remarkable consistency: reward CV of only 0.26%, slope std of 0.01%, and *baimu fang* count identically +5 across all seeds. Township B also shows excellent stability with reward CV of 5.9% and slope std of 0.04%. Across all three townships, the block-level optimization landscape appears to have well-defined basins of attraction: no seed in any township–seed combination produced a degenerate or bimodal trajectory under the tested settings.

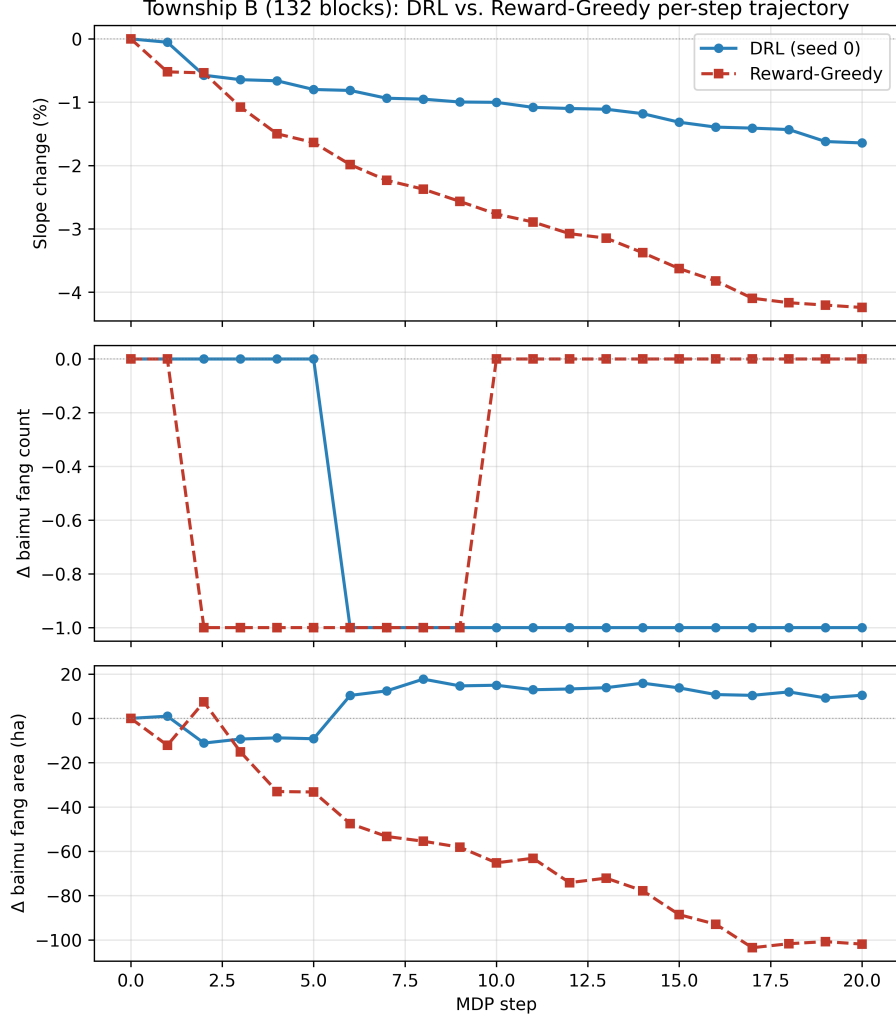


Figure 5: Township B (132 blocks): per-step trajectory comparison between DRL (seed 0, blue) and the Reward-Greedy one-step lookahead baseline (red) that shares DRL’s reward function and weights. Top: cumulative slope change. Middle: *baimu fang* count change. Bottom: *baimu fang* area change. Both methods share the same connectivity-aware engine and total swap budget. Reward-Greedy attains the largest slope reduction in the table by greedily removing high-slope farmland parcels at every step, but those parcels are predominantly in or near high-area qualifying fields, so its *baimu fang* area collapses to -101.9 ha. DRL trades ~ 2.6 percentage points of slope reduction to retain and slightly expand qualifying area, ending at $+10.5$ ha (seed 0). This is the strongest evidence in the paper that, under these reward weights and township geometry, the trained policy can deliver a multi-step advantage on top of access to the multi-objective signal.

Table 13: Per-seed DRL results across all three townships. Std row shows standard deviation. CV row shows coefficient of variation of total reward.

Township	Seed	Slope (%)	Cont.	Baimu cnt	Reward
A	0	-2.67	+0.149	0	154.1
	1	-1.78	+0.154	0	140.3
	2	-2.92	+0.092	0	161.2
	3	-2.92	+0.140	0	159.8
	4	-2.74	+0.113	0	152.7
	<i>CV</i>				<i>5.3%</i>
B	0	-1.64	+0.115	-1	37.6
	1	-1.56	+0.127	-1	38.7
	2	-1.63	+0.115	-1	42.0
	3	-1.66	+0.115	-1	40.5
	4	-1.69	+0.100	-1	35.6
	<i>CV</i>				<i>5.9%</i>
C	0	-0.32	+0.018	+5	265.8
	1	-0.35	+0.027	+5	267.0
	2	-0.33	+0.024	+5	266.6
	3	-0.34	+0.021	+5	265.2
	4	-0.33	+0.025	+5	266.3
	<i>CV</i>				<i>0.26%</i>

5.6. Multi-objective Pareto analysis

Fig. 6 presents the multi-objective Pareto analysis across all methods including the Reward-Greedy one-step lookahead baseline. In the slope-*baimu fang* count space (Fig. 7), Reward-Greedy achieves non-negative *baimu fang* count change on all three townships while maintaining negative slope change, and DRL does so on Townships A and C. On Township B, DRL loses one qualifying field but is the only method with a positive *baimu fang* area outcome. This distinction matters for policy interpretation: count and area are related but not interchangeable indicators.

On Township A, both DRL and Reward-Greedy lie on the Pareto front between Greedy-Global (best slope, worst *baimu fang*) and Round-Robin (worst slope, best *baimu fang* area), with Reward-Greedy slightly closer to the slope axis. The most interpretable summary of the cross-township pattern is: *block-level abstraction plus a reward-aware decision rule (myopic or learned) defines the multi-objective Pareto frontier; learned multi-step planning on top of that contributes a township-specific increment that is most pronounced on Township B.*

On Township C, Round-Robin achieves the largest *baimu fang* area increase (+114.4 ha vs. DRL’s +27.7 ha and Reward-Greedy’s +28.0 ha), but at the cost of an actual slope *increase* (+1.10%); practitioners interested in maximizing total consolidated area on this township may prefer Round-Robin, while those interested in increasing the count of qualifying fields without raising mean farmland slope will prefer Reward-Greedy or DRL.

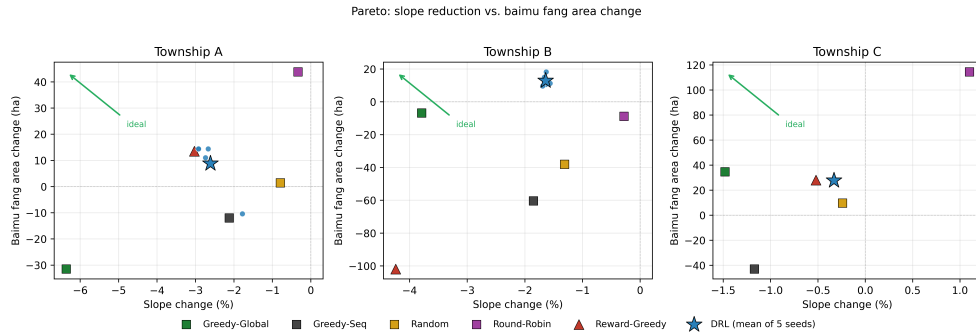


Figure 6: Multi-objective Pareto analysis: slope reduction vs. *baimu fang* area change across all three townships. Baseline methods shown as squares; DRL seeds as blue circles; DRL mean as blue star; Reward-Greedy as red triangle. Green arrows indicate the ideal direction (more slope reduction and more *baimu fang* area). DRL is the only method on Township B that achieves a positive *baimu fang* area change.

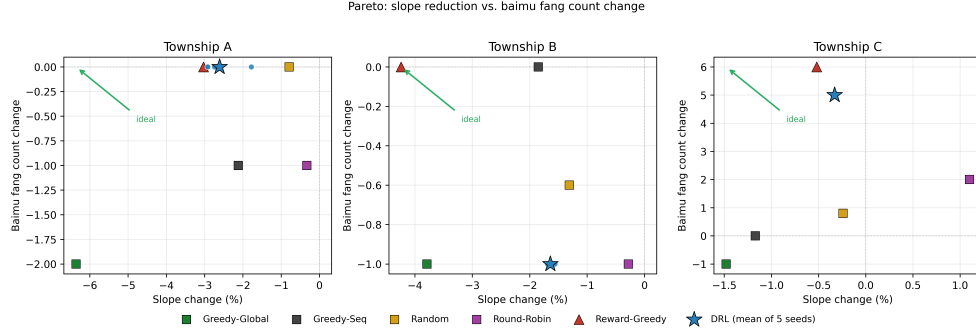


Figure 7: Pareto analysis: slope reduction vs. *baimu fang* count change. Reward-Greedy (reported in Table 7: +0, +0, +6 across A/B/C) achieves non-negative count change on all three townships while maintaining negative slope. DRL achieves this on Townships A and C but loses one qualifying field on Township B, where its advantage lies instead in positive *baimu fang* area. The reward-unaware baselines fail at least one of the two indicators on at least one township.

5.7. Sensitivity to the swaps-per-step parameter κ

The MDP horizon $H = B_{\max}/\kappa$ is the parameter that mediates between “many small block-level decisions” (κ small, H large, regressing toward parcel-level behaviour) and “few large block-level decisions” (κ large, H small, leaving the agent without enough macro-level steps to learn an interesting strategy). Section 4.3 argued for $\kappa = 5$ on heuristic grounds. We now report a small-scale ablation that quantifies the sensitivity directly. We re-train the same MaskablePPO architecture on Township A (the cheapest township) for $\kappa \in \{1, 3, 5, 10, 20\}$, holding the total swap budget at $B_{\max} = 100$ and using a 50,000-timestep budget per setting; all other hyperparameters and the random seed are identical. Results are reported in Table 14.

The behaviour is non-monotone and follows the qualitative prediction in Section 4.3. The two extremes both fail. At $\kappa = 1$ the agent makes 100 near-independent block selections and ends with reward -47.9 , slope reduction of only -0.20% , and—most diagnostically—commits only 8/100 paired swaps, because most of its block selections trigger a within-block engine that can find no slope-improving paired swap and so consumes no budget. This is proxy evidence consistent with the near-independent regime predicted by the geometric analysis, but it does not replace a direct parcel-level DRL benchmark. At $\kappa = 20$ the agent makes only $H = 5$ macroscopic decisions per episode, a horizon too short for PPO to learn a meaningful per-block scoring

Table 14: Sensitivity of the block-level DRL agent to the swaps-per-step parameter κ on Township A (78 blocks). The total paired-swap budget is fixed at 100; the MDP horizon is $H = 100/\kappa$. Each κ is trained for 50,000 timesteps with seed 0 on CPU and evaluated deterministically. “Budget used” is the number of paired swaps actually committed in the deterministic episode out of the 100-swap cap. Reward is the cumulative episodic return under the multi-component reward of Eq. 7 with the same weights used elsewhere in the paper. The $\kappa = 5$ value used in the main experiments achieves the highest cumulative reward (bold).

κ	H	Slope (%)	Cont.	Baimu cnt	Baimu ha	Budget used	Reward
1	100	−0.20	−0.011	+2	+6.7	8/100	−47.9
3	33	−2.07	+0.113	+1	−4.5	42/100	78.5
5	20	−2.01	+0.074	+1	+18.4	38/100	85.5
10	10	−0.59	+0.009	+1	+22.0	18/100	52.2
20	5	−0.06	+0.002	+1	−1.7	4/100	17.0

policy from the Township-A reward signal; the trained policy ends with reward 17.0, slope reduction of -0.06% , and a negligible budget commitment of 4/100.

The middle of the range gives the strongest results. $\kappa = 3$ ($H = 33$) and $\kappa = 5$ ($H = 20$) both reach a slope reduction of about -2% on Township A, with $\kappa = 5$ achieving the highest cumulative reward (85.5) and the most positive *baimu fang* area outcome (+18.4 ha). $\kappa = 10$ ($H = 10$) trails on slope (-0.59%) but is the strongest on *baimu fang* area (+22.0 ha), suggesting that for installations whose policy priority is total consolidated-area growth rather than slope reduction, a larger κ may be preferable. The default $\kappa = 5$ used throughout this paper is therefore not a universally optimal choice but a balanced one; practitioners with a different multi-objective preference can re-tune κ in the same range and inherit the framework’s qualitative behaviour.

Because Township B is the critical case for the learned scheduler, we also ran a short-training diagnostic on that township for $\kappa \in \{3, 5, 10\}$ using seed 0 and 50,000 timesteps (Table 15). This diagnostic is not directly comparable to the main 200,000-timestep, five-seed Township B experiment: all three short-training policies commit only 7–13 of the 100 possible swaps, and none reproduces the main trained policy’s positive, stable Township B outcome. Its value is therefore cautionary. It shows that κ interacts with training budget and township geometry, and that the main Township B claim should be tied to the reported training protocol rather than generalized to arbitrary

horizons or short training runs.

Table 15: Sensitivity of the block-level DRL agent to the swaps-per-step parameter κ on Township B (132 blocks). The total paired-swap budget is fixed at 100; the MDP horizon is $H = 100/\kappa$. Each κ is trained for 50,000 timesteps with seed 0 on CPU and evaluated deterministically. “Budget used” is the number of paired swaps actually committed in the deterministic episode out of the 100-swap cap.

κ	H	Slope (%)	Cont.	Baimu cnt	Baimu ha	Budget used	Reward
3	33	−0.10	+0.026	−1	+16.5	7/100	−2.3
5	20	−0.66	+0.021	−1	−3.4	13/100	−0.6
10	10	−0.30	+0.028	−1	+14.0	8/100	3.8

6. Discussion

6.1. When block-level learned scheduling adds value

The results suggest that the main value of the proposed framework is not that DRL universally dominates simpler planning rules. Its value is conditional: block-level abstraction first creates a policy-readable scenario space, and learned scheduling then helps when short-horizon reward maximization creates an undesirable trade-off. Township B is the clearest example, because Reward-Greedy and finite-depth lookahead pursue strong slope reduction while dismantling high-area qualifying field blocks, whereas the learned scheduler preserves more consolidated-field area at the cost of weaker slope reduction.

Three structures make this conditional scheduling problem visible:

Resource competition. With a fixed budget of 100 swaps and $\kappa = 5$ swaps per step, the agent has exactly 20 block selections to make. Each selection consumes 5% of the total budget; investing in block b means *not* investing in block b' . This creates a genuine allocation problem where the ordering and selection of blocks affects the final outcome.

Spatial spillover. The connectivity-aware greedy engine ($\gamma = 1.0$ for forest conversion) creates cross-block dependencies: converting forest to farmland at a block boundary increases the farmland neighbor count of parcels in adjacent blocks, making them more attractive targets. This spillover effect means that investing in block b at step t changes the expected reward from block b' at step $t + 1$ —precisely the inter-step dependency that DRL exploits.

Baimu fang threshold effects. The ≥ 6.67 ha threshold for *baimu fang* qualification creates discontinuous reward dynamics: a farmland cluster just below the threshold yields zero *baimu fang* reward, but one additional parcel conversion can push it above the threshold, generating a large positive signal. Strategic sequencing can therefore matter when blocks differ in whether they protect existing qualifying area or create new near-threshold opportunities.

Scope of the DRL claim. These mechanisms explain why block-level abstraction can create a useful sequential decision problem, but they do not prove that every parcel-level DRL formulation must fail, nor that RL is uniquely necessary. The present evidence is positive rather than exhaustive: it shows that the proposed block-level formulation is trainable, interpretable, and competitive under matched within-block execution. The $\kappa = 1$ experiment provides proxy evidence for a near-independent regime, and the Township-B short-horizon checks show that depth-2 and depth-3 reward search do not reproduce the DRL *baimu fang* area outcome. A full parcel-level DRL baseline and longer-horizon deterministic planning remain separate robustness tests.

6.2. The “invest-then-recover” strategy on Township C

The DRL agent’s behavior on Township C merits detailed analysis. Examining the block-by-block trajectory recorded in the per-step evaluation log (Fig. 8; full per-step CSV provided as Supplementary Material), the agent follows a two-phase strategy:

Phase 1 (steps 1–10): *Baimu fang* investment. The agent selects blocks at the boundaries of existing farmland clusters, converting forest to farmland to push near-threshold clusters across the 6.67 ha line. The cost is paid up front: the very first block selection raises the area-weighted mean farmland slope by $\approx +0.93\%$ over the initial value, because the agent is converting gentle-slope forest (whose removal as future farmland would have reduced slope) to farmland *now*, and is removing relatively steep farmland *later*. From step 2 onward, the slope penalty monotonically decreases as the engine starts pairing each new forest→farmland conversion with a higher-slope farmland→forest removal. By the end of step 10, all 5 new *baimu fang* have been formed (count goes $0 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow 4 \rightarrow 4 \rightarrow 5$ at steps 1, $\dots$, 10).

Phase 2 (steps 11–20): Slope recovery. With all five *baimu fang* already formed, the agent switches to blocks whose within-block engine yields large slope improvements. The *baimu fang* count stays exactly at +5 from

step 10 onward (the agent does not form new fields, but importantly, it also does not lose any), and the slope change crosses zero around step 14 and continues decreasing to a final value of -0.32% by step 20.

This temporal strategy is impossible for any *slope-only* single-step greedy method to discover: Greedy-Global would never select a block that worsens slope in the short term, even if doing so enables superior *baimu fang* formation in the long term. We note, however, that a *reward-aware* one-step lookahead (Reward-Greedy, Section 4.7) does discover an analogous count-investing pattern on this township purely through one-step reward maximization—it forms +6 new *baimu fang*, even more than DRL’s +5. The clearest evidence that DRL provides a genuinely *multi-step* advantage rather than just access to the multi-objective signal therefore comes not from Township C, but from Township B (Section 5.2), where Reward-Greedy’s myopic preference for the slope channel loses ~ 115 ha of *baimu fang* area relative to DRL. The agent’s plateau at +5 from step 10 onward on Township C nonetheless illustrates a clean form of objective switching: once the agent identifies that no further qualifying field can be formed within the remaining budget, it switches its effective objective to the slope channel of the reward, which is exactly the kind of value-function-driven behavior that block-level abstraction is meant to enable.

6.3. Policy implications

The block-level DRL framework suggests several implications for farmland consolidation policy and land-resource decision support:

Use at the project-bank and scenario-comparison stages. Consolidation projects are administered through project-bank screening, candidate-area comparison, budget allocation, field investigation, village negotiation, and formal approval. The block-level agent approximates only the early screening and comparison part of this workflow. Its sequence of block selections can be translated into planning language, such as “prioritize consolidation in the northern valley cluster, then the eastern hillside,” but that sequence still requires field verification and administrative review. The most defensible use is to rank candidate blocks, compare objective trade-offs, and identify where planners should investigate further.

Threshold-based field-block metrics as decision criteria. Current practice relies on expert judgment to identify *baimu fang* formation opportunities. The block-level abstraction in our framework—used by either DRL or the Reward-Greedy lookahead—automates this assessment, learning or

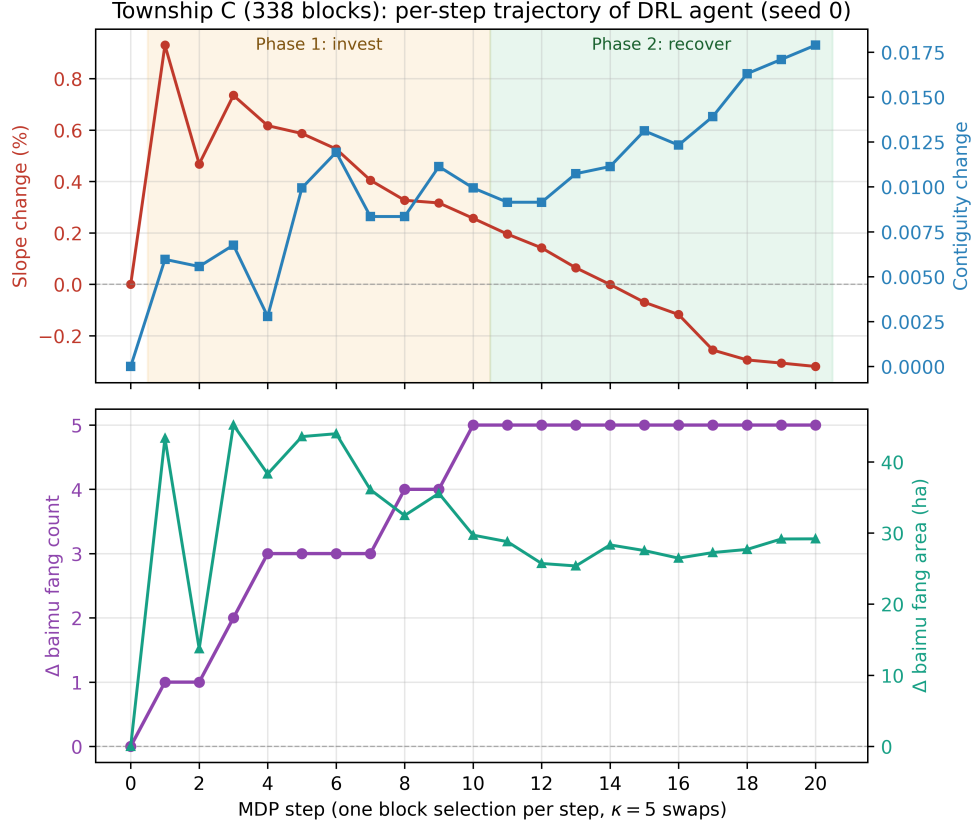


Figure 8: Township C (338 blocks) per-step trajectory of the DRL agent (seed 0, deterministic evaluation). Top panel: cumulative slope change (%) and contiguity change. Bottom panel: *baimu fang* count change and *baimu fang* area change (ha). The two phases are separated by a vertical shading break at step 10. All five new *baimu fang* are formed during Phase 1 (cumulative count: $0 \rightarrow 5$) at the cost of an initial slope penalty of $\approx +0.93\%$; in Phase 2 the count is held fixed at $+5$ and the slope penalty is recovered, ending at -0.32% . The other four seeds (1–4) follow the same two-phase pattern with very low variance, consistent with the per-seed reward CV of 0.26% reported in Table 13; their trajectories are provided as Supplementary Material.

computing from the spatial configuration which blocks offer the best consolidation potential. On Township C, both DRL (+5 new fields) and Reward-Greedy (+6) far exceed the four reward-unaware baselines (at most +2), suggesting that block-level multi-objective evaluation can identify consolidation opportunities that simple slope-driven heuristics miss. Outside China, the same design can substitute locally meaningful thresholds, such as minimum mechanized-field size, contiguous management-unit area, or restoration-patch size.

Multi-objective balancing rather than automatic optimization. Different townships may have different policy priorities: some emphasize slope reduction (for erosion prevention), others prioritize block formation (for mechanization), and others may prioritize exact area balance or ownership feasibility. The reward weights in Table 4 should therefore be treated as one scenario setting. Local planning agencies would need to re-run the framework under their preferred weights and constraints, then compare the resulting trade-off tables rather than accept one reward configuration as normative.

Property rights, social negotiation, and the optimizer’s role. A critical consideration for real-world implementation is that cadastral parcels carry complex property rights, including the *contracted management rights* (*chengbao jingyingquan*) attached to rural households and village cooperative units [17]. Farmland–forest exchanges across these boundaries incur transaction costs and require voluntary participation, compensation design, and administrative approval. We therefore stress that the block-level DRL framework produces a physical scenario frontier under stated spatial objectives, not a binding administrative prescription. It is intended to surface promising candidate areas and trade-offs, leaving ownership verification, consent, timing, and compensation to local planning institutions.

The block construction method can support this governance boundary, but it does not yet encode ownership. Natural-barrier segmentation and constrained clustering create physically coherent blocks, but we do not observe household, village-group, or cooperative boundaries in the optimization state. A practitioner who wants to minimize negotiation friction should pass those administrative or ownership polygons into the block construction step as additional hard barriers. That extension would make the decision units socially meaningful as well as physically coherent, but it remains outside the present experiment.

International transferability. The named *baimu fang* target is specific to the Chinese high-standard farmland context, but the modelling role

of the metric is more general. It represents a minimum contiguous field-block threshold. Other land-use systems could substitute locally relevant thresholds, such as minimum mechanized-field area, minimum contiguous management-unit area, or restoration-patch size. The transferable part of the framework is therefore not the 6.67 ha value itself, but the coupling of a threshold-based planning metric with block-level scenario comparison.

6.4. Limitations

Several limitations should be acknowledged.

Parcel-count conservation does not imply farmland-area conservation. This is the most important technical boundary for policy use. The connectivity-aware greedy engine pairs each farmland removal with a forest conversion, exactly conserving the parcel *count* of each land-use class by construction. However, paired parcels in real cadastral data have heterogeneous areas. Across an episode of 100 paired swaps, the cumulative net change in total farmland *area* can therefore be non-zero, with implications for the area-weighted mean slope (Eq. 1) and for compliance with China’s farmland *occupation–compensation balance* (*zhanbu pingheng*) policy. The area-drift audit in Section 5.3 shows that the DRL scenarios have small average net drift in the original engine, but the Township-B area-tolerance stress test also shows that replaying the learned histories under stricter area and slope checks loses the positive *baimu fang* area result. This means the current learned policies are not implementation-ready under hard area-balance rules. A next iteration should train the agent directly in an area-constrained transition environment, or formulate farmland area as a hard constrained-MDP condition rather than as a post-hoc diagnostic.

No direct parcel-level DRL baseline. The motivation for block-level abstraction rests on the near-independence diagnosis and on the $\kappa = 1$ proxy ablation, but this manuscript does not train a full policy that selects among individual parcels in the original cadastral action space. This limits the strength of any negative claim about parcel-level DRL. The established result is narrower: the block-level formulation is trainable and adds value in some multi-objective tension cases under the tested settings. The new Township-B depth-2 and depth-3 lookahead checks strengthen the comparison against short-horizon block-level planning, but they do not replace a direct parcel-level PPO or parcel-level reward-greedy baseline.

Morphological simplification of *baimu fang* definition. The current BFS-based identification of *baimu fang* considers only topological con-

nectivity and aggregate area (≥ 6.67 ha), without imposing shape constraints. The compactness audit in Section 5.3 shows that this is not a minor issue: no initial or DRL-final component reaches $IPQ \geq 0.20$ under the Queen-contiguity definition. Future work should incorporate compactness metrics—such as the isoperimetric quotient ($4\pi A/P^2$), minimum bounding rectangle ratio, or convexity index—as qualification criteria and as reward components.

Slope-*baimu fang* trade-off. On Township B, the DRL agent trades slope performance (-1.64% vs. Greedy-Global’s -3.79%) for substantially superior contiguity and *baimu fang* area outcomes. On Township C, the agent trades slope for *baimu fang* count formation. While this reflects rational multi-objective optimization, DRL does not dominate Greedy-Global on slope reduction—a metric that may be the primary concern in some policy contexts. The reward weight configuration determines the trade-off point on the Pareto front; practitioners can adjust weights to emphasize their priority objective.

Block definition sensitivity. The block construction method (barrier segmentation + agglomerative clustering) involves parameter choices (minimum block size 3 parcels, minimum block area 0.5 ha, target ~ 20 parcels per block, large-component threshold 30 parcels). We have not conducted a systematic ablation of these parameters; the consistency of training stability across three townships with different spatial structures suggests reasonable robustness, but a quantitative ablation over target $\in \{10, 20, 40\}$ parcels/block and minimum area $\in \{0.2, 0.5, 1.0\}$ ha is needed to establish whether downstream multi-objective performance is similarly stable.

Greedy-engine connectivity parameters. The within-block scoring uses fixed connectivity weights $\gamma = 1.0$ (forest-to-farmland conversion bonus) and $\delta = 0.5$ (farmland removal penalty). The 2:1 asymmetry encodes a deliberate “active consolidation” bias: rewarding the formation of new contiguous farmland twice as much as preserving existing edges, so that the engine is willing to fill in interior forest gaps rather than only protecting current boundaries. The values were chosen by domain knowledge rather than through optimization; their effect on the slope/contiguity/*baimu fang* trade-off has not been ablated, and different values might yield different strategy profiles. Because all baselines share the same engine, the comparison among methods is internally consistent, but absolute numbers are conditional on this (γ, δ) choice.

Swaps-per-step. The parameter $\kappa = 5$ used throughout the main experiments was chosen heuristically (Section 4.3); the Township-A ablation and

Township-B short-training diagnostic (Section 5.7) show that performance is sensitive to this horizon setting. The Township-B diagnostic is especially cautionary because 50,000-timestep policies under $\kappa \in \{3, 5, 10\}$ commit only a small share of the 100-swap budget. A full grid over κ , reward weights, all three townships, and multiple seeds remains needed.

Reward-Greedy vs. DRL gap is small on A and C. The Reward-Greedy one-step lookahead baseline that shares DRL’s reward function and weights (Table 7) is competitive with DRL on Townships A and C. It exceeds DRL’s slope reduction, contiguity, and *baimu fang* count on Township C, and is slightly better than DRL on Township A on three out of four metrics. The DRL-specific advantage is concentrated on Township B, where Reward-Greedy, depth-2 exhaustive lookahead, and depth-3 beam lookahead all retain strong slope reduction but lose more than 100 ha of *baimu fang* area. We therefore frame the contribution carefully: the block-level abstraction (Section 4.2) and the connectivity-aware engine (Section 4.3.3) carry most of the multi-objective gain, while the trained DRL policy adds a township-dependent increment that is large on B and small on A and C. Deeper value-network-guided rollouts or mixed-integer local search might still reproduce the Township-B effect without RL training, but they remain outside the present comparison.

Single district and statistical scope. All three townships are within Bishan District, limiting geographic generalizability; the framework should be validated on regions with different terrain, climate, and land-use patterns. Statistical reporting is also limited: DRL and Random use 5 seeds, while the deterministic baselines (Greedy-Global, Greedy-Sequential, Round-Robin) report a single trajectory each, so we have not computed confidence intervals or significance tests for the DRL-vs-deterministic-baseline comparisons. Future work should report bootstrap CIs over DRL/Random seeds and, where applicable, perturb deterministic baselines by re-running them with random tie-breaking.

Computational cost. The block-level approach requires GPU training: 200,000 timesteps per seed, ~ 25 minutes (Township A), ~ 50 minutes (B), and ~ 2 hours (C) on a single NVIDIA A100 40GB. By contrast, Greedy-Global produces its result in under 0.01 seconds, and Reward-Greedy in under 5 seconds (deterministic, no training needed). The DRL approach is therefore justified only when the multi-objective optimization capability, particularly the Township-B retention of consolidated-field area that short-horizon reward search fails to reproduce (Section 5.2), is operationally valued.

7. Conclusions

This paper proposed a policy-relevant block abstraction for sequential farmland-consolidation scenario screening. The method is motivated by the gap between parcel-level optimization and the project-area scale at which consolidation planning, budget allocation, field investigation, and negotiation usually occur. Through experiments on three townships spanning 3,607 to 12,950 parcels (78 to 338 blocks) of Third National Land Survey cadastral data, we established the following findings:

1. **Block-level DRL trains stably under the tested settings.** All 15 seed-township combinations reach stable reward plateaus and non-degenerate deterministic trajectories under the operational convergence definition used here. On the largest township (12,950 parcels, 338 blocks), the across-seed reward coefficient of variation is below 1%, providing a stable basis for multi-objective evaluation.
2. **The “macroscopic planning, microscopic execution” paradigm creates observable sequential dependencies under the tested settings.** Resource competition between blocks, spatial spillover via the connectivity-aware greedy engine, and *baimu fang* threshold effects combine to create inter-step dependencies that DRL can exploit in selected cases.
3. **The learned scheduler produces township-specific block sequences.** Under identical hyperparameters, the agent develops balanced, contiguity-focused, and *baimu fang* count-focused trajectories on different townships, reflecting local spatial structure without manual per-township rule design.
4. **The multi-objective gains are driven mostly by block-level abstraction itself, with DRL adding a township-specific increment.** A reward-aware one-step lookahead baseline (Reward-Greedy) that shares DRL’s reward function is competitive with the trained DRL agent on Townships A and C and even slightly exceeds it on the *baimu fang* count metric on Township C (+6 vs. +5). DRL’s distinctive contribution is concentrated on Township B, where it is the only tested method to achieve positive *baimu fang* area growth (+12.9 ha vs. Reward-Greedy’s −101.9 ha, depth-2 lookahead’s −108.8 ha, and depth-3 beam lookahead’s −103.8 ha), avoiding the short-horizon trap of dismantling consolidated fields to chase slope reduction.

5. **The framework aligns with consolidation policy, but only as a screening tool.** Block-level decision-making mirrors early-stage project-bank screening and candidate-area comparison, and *baimu fang* formation is a directly policy-relevant metric. Area-drift auditing shows small DRL net farmland-area changes under the original engine, but the area-tolerance stress test changes the Township-B outcome, and the compactness audit shows that raw topological field counts are upper-bound indicators. The framework is therefore best interpreted as a pre-screening and scenario-exploration tool rather than an implementable land-adjustment plan.

These results indicate that spatial abstraction, lifting the optimization from individual parcels to policy-relevant consolidation units, is a defensible design choice for land-use scenario screening. The framework’s most defensible contribution is the joint package of block-level abstraction, the connectivity-aware engine, and a learnable scheduler whose specific advantage appears in multi-objective tension cases such as Township B. The next robustness priorities are direct parcel-level baselines, fully trained area-constrained policies, shape-aware field-block definitions, block-granularity ablation, cross-district validation, integration with GIS planning tools, and extension to multi-land-type consolidation scenarios involving construction land reclamation and ecological restoration.

CRedit authorship contribution statement

[Anonymous Author]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The replication code for the block-level environment, training scripts, baselines, evaluation utilities, and figure-generation pipelines, as well as anonymized aggregate result files supporting all tables and figures in this paper, are available for double-anonymized review at <https://anonymous.4open.science/r/block-level-farmland-drl-8456/>. The full Third National Land Survey (TNLS) cadastral dataset for Bishan District is governed by the access regulations of the relevant Chinese authorities and cannot be redistributed by the author. Readers wishing to reproduce the experiments on the original data should obtain access through the proper channels of China’s Ministry of Natural Resources, after which the present codebase can ingest the data via the documented preprocessing pipeline. Synthetic test fixtures that exercise every code path are included in the repository so that the methodology can be validated end-to-end without access to the restricted dataset. Slope features were derived from the publicly available Copernicus DEM GLO-30 [5] via standard zonal statistics (mean slope per parcel, computed in the EPSG:32648 projection).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used AI-assisted tools to support language editing, LaTeX formatting, and journal-format compliance checks. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the submitted manuscript.

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